

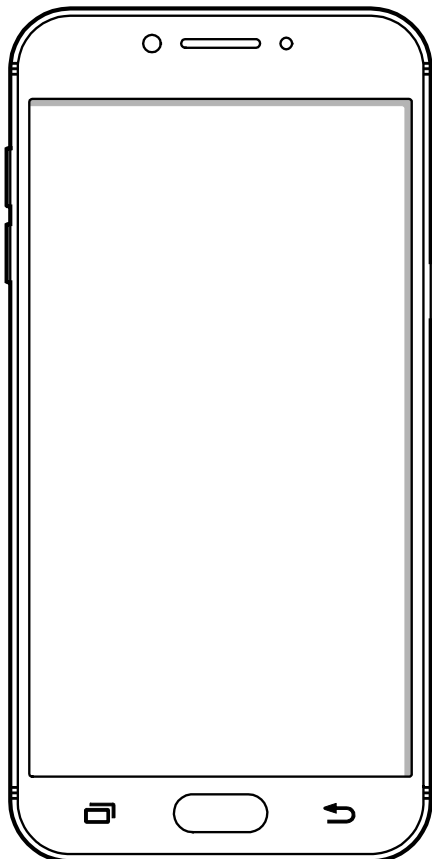
SAMSUNG

Mobile Device
SM-C7010/C7018
Common

SERVICE **Manual**

Mobile Device

CONTENTS

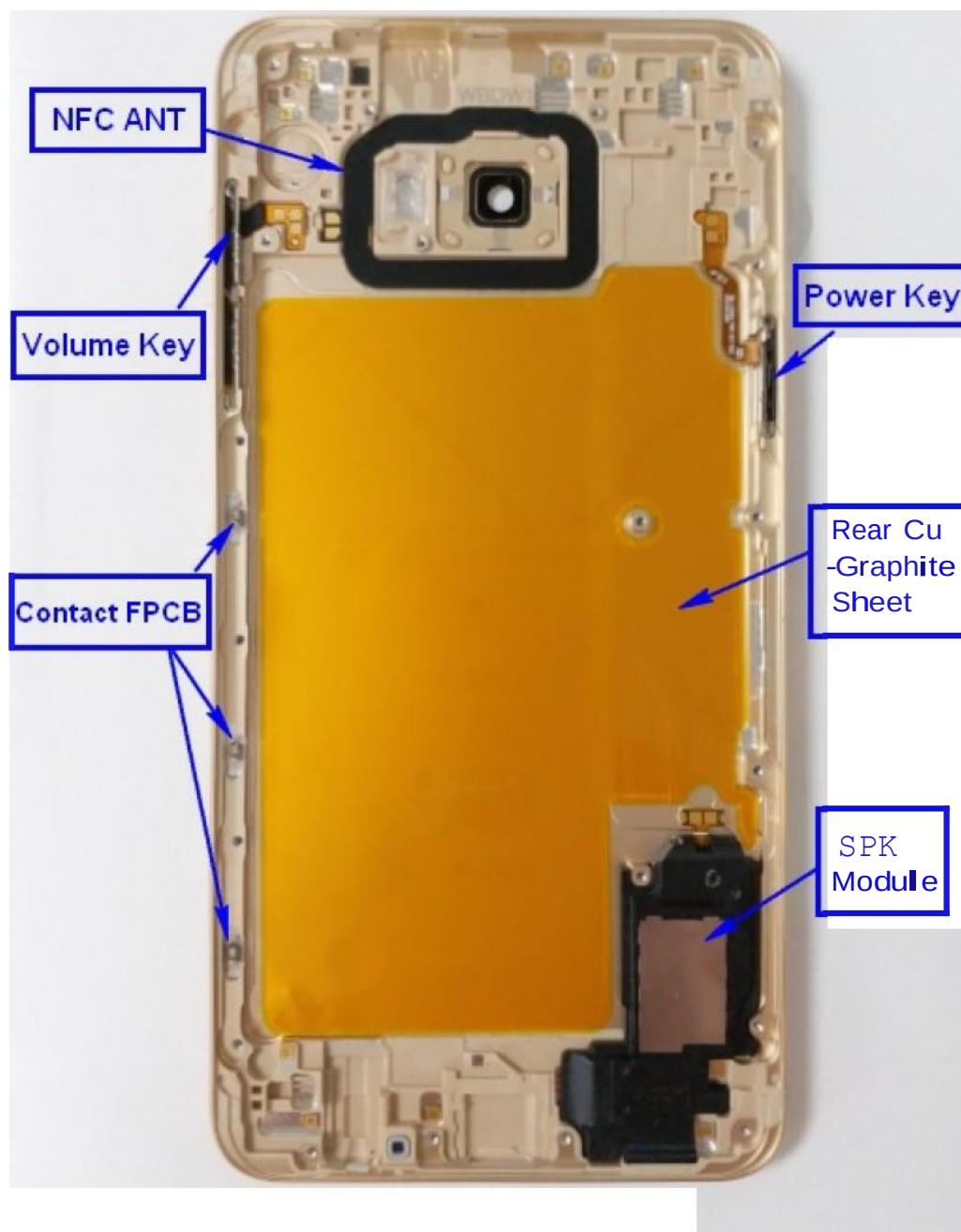


1. Safety Precautions
2. Specification
3. Product Function
4. Exploded View and Parts list
5. MAIN Electrical Parts List
6. Level 1 Repair
7. Level 2 Repair
8. Level 3 Repair
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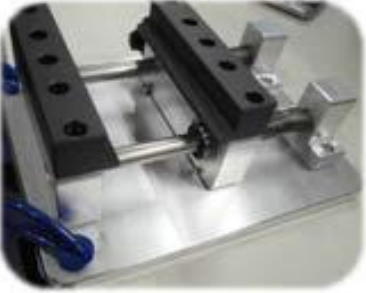
7. Level 2 Repair

7-1. 后壳上相关组件：



7. Level 2 Repair

7-2. 必要工具

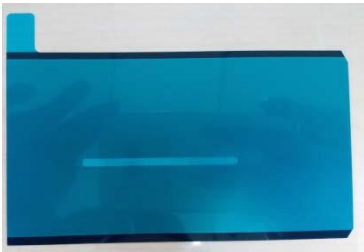
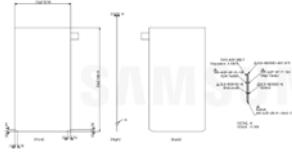
	
镊子/拆卸棒/螺丝刀	防静电手套
	
棉签	酒精
	
OCTA 拆卸架	OCTA 拆卸上部
	
OCTA 安装 JIG(1N 1 分钟)	加热箱

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7. Level 2 Repair

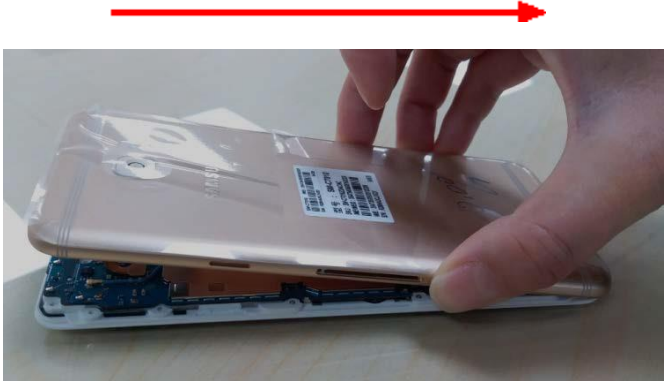
7-3. 维修后需要更换的部分

BOM 描述	图片	备注
A/S-TAPE_ MAIN_WINDOW_SVC [GH02-13819A] [GH02-13818A]	 TOP_GH02-13819A  BOT_GH02-13818A	
A/S-OCTA_CU SHEET_SVC [GH02-14091A]		
A/S-OCTA SCF TAPE [GH81-14369A]	 	维修前部时进行更换
A/S-TAPE HOTMELT-HR_SVC [GH02-13827A]	 	
A/S-DOUBLE TAPE ASSY [GH81-14457A]	 DOUBLE TAPE ASS'Y	

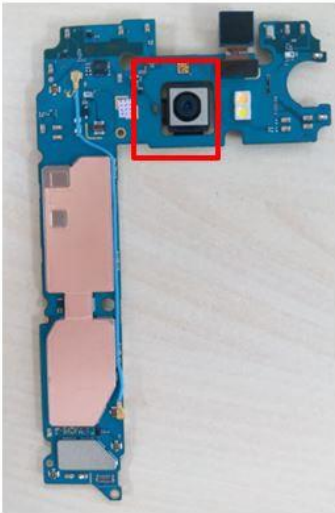
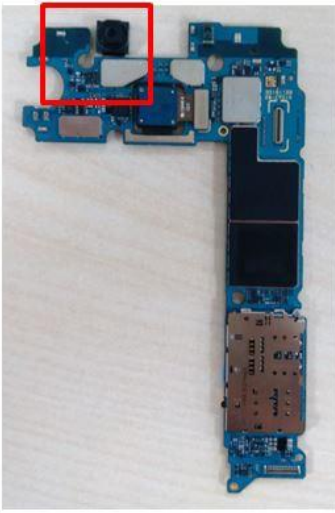
7. Level 2 Repair

7-4. 拆装

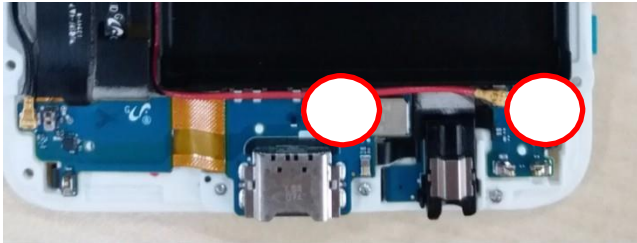
※注意带上防静电手环！

 将设备放入加热箱 (温度：70°±5, 时间：15 分钟)	 当 OCTA 与前部组件分离时，拿出设备
[1] [1] 在 70~80℃温度下，加热 15~30 分钟后，将设备从加热箱拿出。	2] 通过使用后壳拆卸工具将后壳从设备分离。
	
[3] 拧开 22 个螺丝。 ※ 注意 注意擦伤或损坏	[4] 从顶部到底部将后盖从设备边框分离 ※ 注意 注意损坏

7. Level 2 Repair

	
[5] 分离电池连接器	[6] 1) 分离主屏幕键连接器 2) 使用拆卸帮拆除同轴线，副 PBA 连接器，将 PBA 从边框分离 ※ 注意 1) 注意损坏。 2) 拆除 PBA 垫片后，拆除 PBA. 3) 使用拆卸棒
 	
[7] 将前置后置摄像头从 PBA 分离 ※ 注意 小心不要损坏	[8] 分离接收器，马达，电池。

7. Level 2 Repair



[9]

- 1) 松开 2 处螺丝.
- 2) 将触摸键和副 PBA 从边框分离

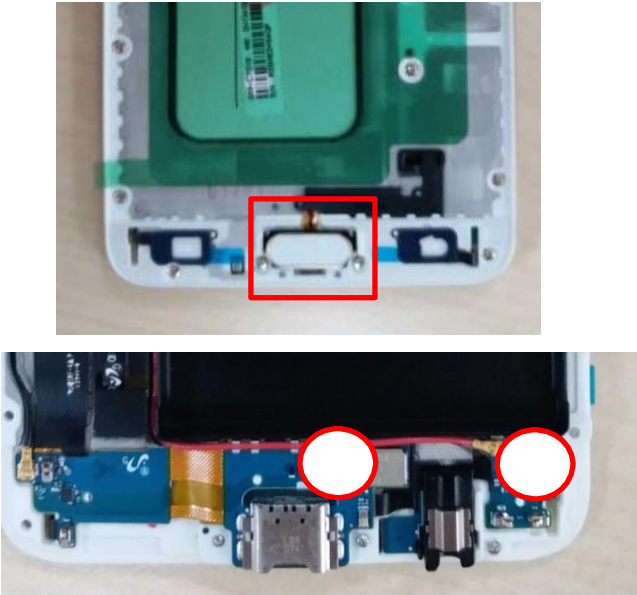
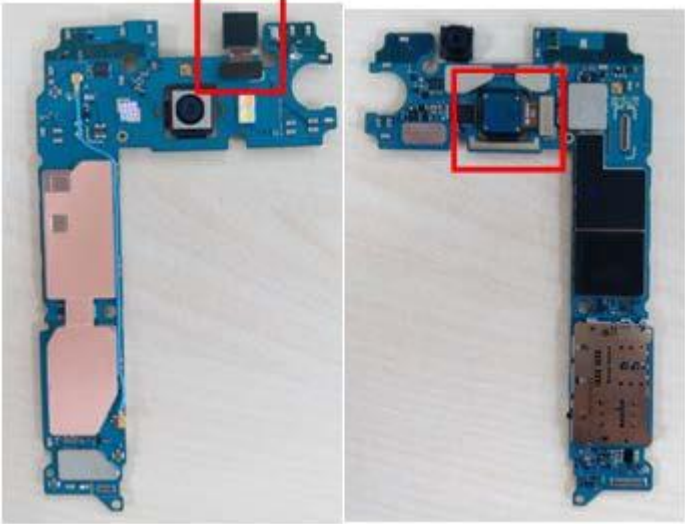
※ 注意

小心不要损坏

7. Level 2 Repair

7-5. 组装

※ 注意带上防静电手环！

	
[1] 1) 组装主屏幕键 2) 组装副 PBA 并拧 1 处螺钉	[2] 1) 组装马达和接收器 2) 使用电池按压工具按压电池
	
[3] 1) 组装前置和后置摄像头 ※ 注意 小心不要造成损坏	[4] 1) 组装副 PBA 连接器。 2) 将 PBA 组件安装到支架上。 3) 组装电池连接器

7. Level 2 Repair

	
[5] 组装主屏幕键连接器。 ※ 注意 小心不要造成损坏	[6] 组装同轴电缆
	
[7] 1) 将后部组装到支架组件上(从下到上)	[8] 拧紧 22 处螺钉

7. Level 2 Repair



[9] 分离支架上的胶贴。

[10]
1) 组 装 OCTA 连接器。
2) 组装 OCTA。



[11] 使用 OCTA 按下工具按压设备 (1 分钟 1N)

4. Exploded View and Parts List

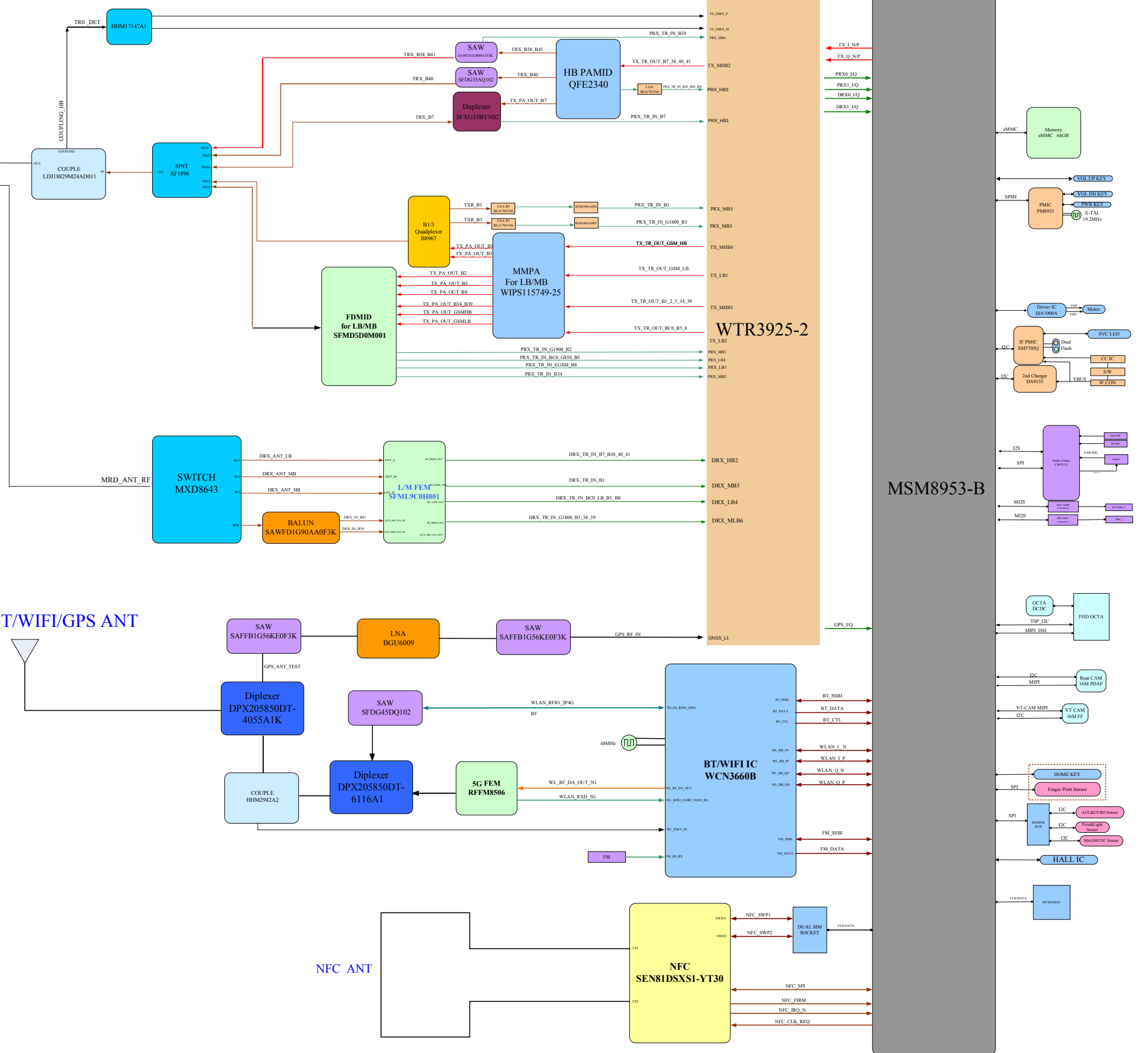
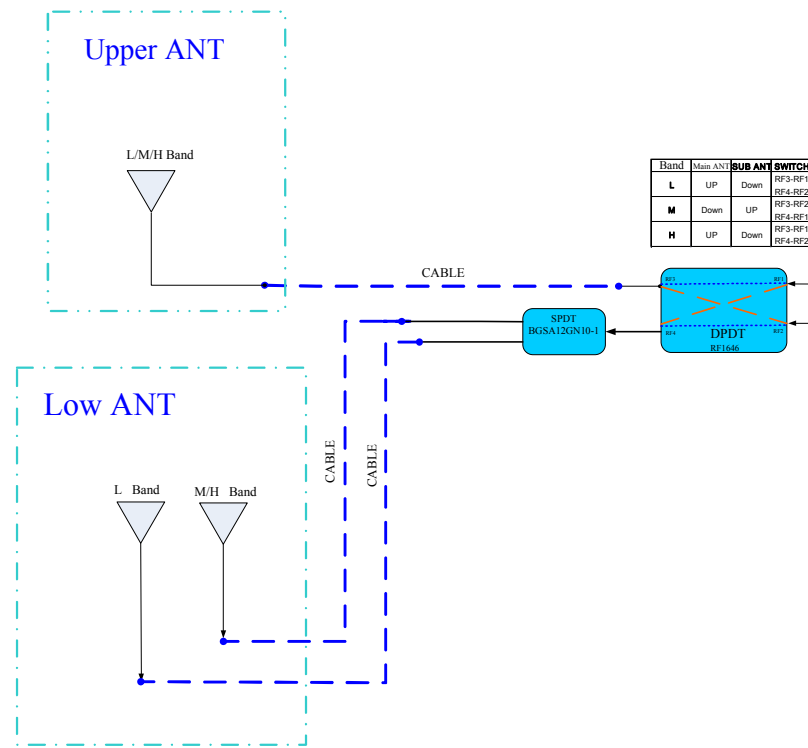
4-2. Cellular phone Parts list

Design LOC	Description	SEC CODE	中文补充翻译
QCR12	SCREW-MACHINE	6001-001530	螺丝
QCR152	SCREW-MACHINE	6001-003226	螺丝
QBA01	INCELL BATTERY PACK-EB-BC701ABE,3300,60	GH43-04694A	电池
QCO01	CASE-LCD CONN	GH61-12022A	LCD 垫片
QCB01	CBF COAXIAL CABLE-82.7MM(SM-C7010)	GH39-01909A	同轴线
QCA01	ASSY CAMERA-MAIN_16M_1/3.1"_C5010	GH96-10596A	主摄像头
QVK01	KEY FPCB-POWER(SM-C5000)	GH59-14667A	电源键
QVK02	KEY FPCB-VOLUME(SM-C9000)	GH59-14715A	音量键
QSP01	ASSY SPEAKER-MODULE(SM-C7010)	GH96-10537A	扬声器模块
QRF04	ASSY RUBBER-MIC	GH98-37607A	MIC-橡胶
QSK01	ASSY KEY-SIDE	GH98-40776A	侧键
QAR01	AUDIO-RECEIVER	3009-001716	音频接收器
QCR92	SCREW-MACHINE	6001-002261	螺丝
QMO01	MOTOR LINEAR VIBRATION-SM_G890A	GH31-00728A	马达
QCB02	CBF COAXIAL CABLE-36.7MM (SM-C7010)	GH39-01898A	同轴线
QCB03	CBF COAXIAL CABLE-81.2MM (SM-C7010)	GH39-01901A	同轴线
QRF01	TIM PAD-MAIN CHARGE	GH67-03524A	垫片
QRF02	TIM PAD-AP TAPE	GH67-03591A	垫片
QME01	ASSY SUB PBA-SM-C7010	GH96-10536A	副PBA
QKP01	HOME KEY ASSY-SM-C7010	GH96-10584A	主屏幕键
QRF04	ASSY RUBBER-MIC	GH98-37607A	MIC-橡胶
QSM01	ASSY DECO-SIM TRAY	GH98-41150A	SIM卡托
QCA02	ASSY CAMERA-MODULE_VT_16M_1/3.1"_C5010	GH96-10664A	摄像头
QMP01	SVC PBA-PBA MAIN(CHNA),SM-C7010	GH82-13755A	主板
QFR01	SVC LCD ASSY-OCTA,(E/WHT),SM-C7010	GH97-20149A	LCD屏
QFR02	SVC CASE ASSY-BRK(GOLD)	GH82-13777A	前部支架
QRE01	A/S ASSY-REAR_C7010(ZD)	GH82-13765A	后部支架
QRT01	TAPE PET-OCTA_TOP	GH02-13819A	胶贴
QRT02	TAPE PET-OCTA_BOT	GH02-13818A	胶贴
QRT03	TAPE DOUBLE FACE-HR	GH02-13827A	胶贴
QRT04	A/S-DOUBLE TAPE ASSY	GH81-14457A	胶贴
QRT05	GRAPHITE THERMAL SHEET-OCTA CU	GH02-14091A	胶贴
QRT06	A/S-OCTA SCF TAPE	GH81-14369A	胶贴

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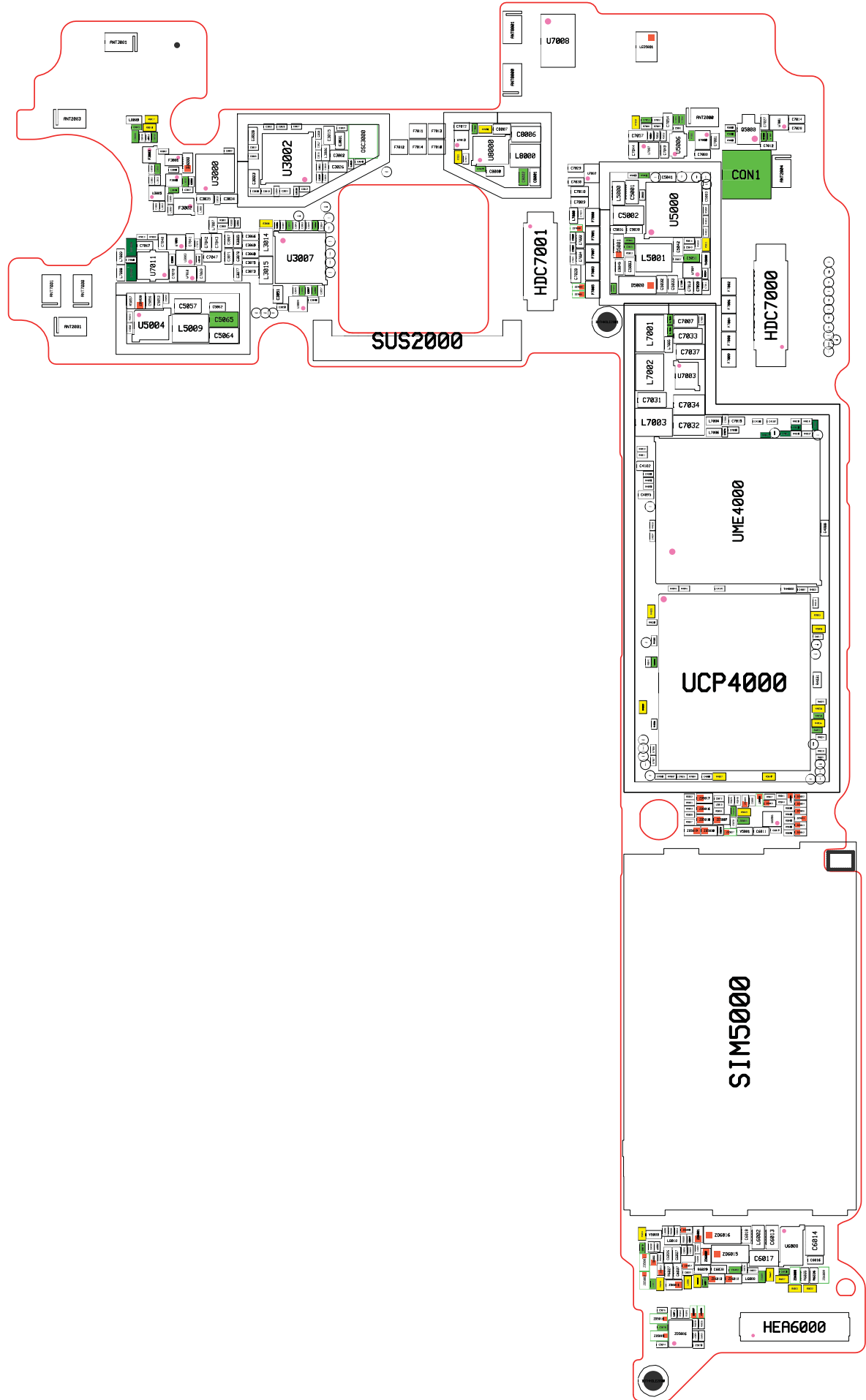
ata from this document without Samsung's written authorization is strictly prohibited.

SM-C7010 SVC Manual Block Diagram



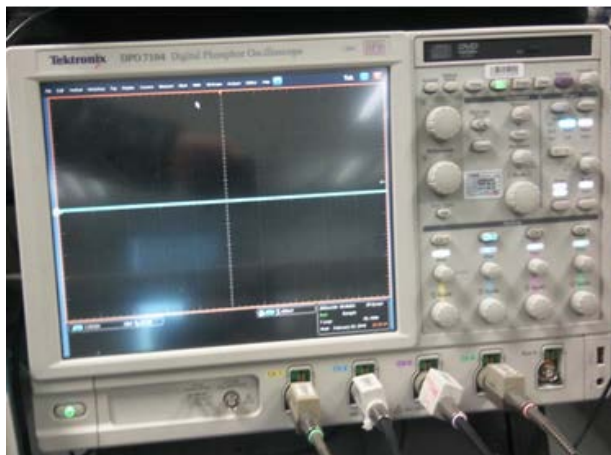
Information of RF Bands supported on SM-C7010 Model

GSM850 : Rx : 869.2 ~ 893.8 MHz / Tx : 824.2 ~ 848.8 MHz
 GSM1900 : Rx : 1930.2 ~ 1989.8 MHz / Tx : 1850 .2 ~ 1909 .8 MHz
 GSM900 : Rx : 925.2 ~ 959.8 MHz / Tx : 880.2 ~ 914.8 MHz
 GSM1800 : Rx : 1805.2 ~ 1879.8 MHz / Tx : 1710 .2 ~ 1784 .8 MH
 WCDMA Band 2 : Rx : 1930 ~ 1990 MHz / Tx : 1850 ~ 1910 MHz
 WCDMA Band 5 : Rx : 871.4 ~ 891.6 MHz / Tx : 826.4 ~ 846.6 MHz
 WCDMA Band1 : Rx : 2112.4 ~ 2167 .6 MHz / Tx : 1922.4 ~ 1977.6 MHz
 WCDMA Band 8 : Rx : 927.4 ~ 957.6 MHz / Tx : 882.4 ~ 912.6 MHz
 CDMA BC0: RX: 869~894MHz / TX: 824~894MHz
 LTE Band 5 : Rx : 869 ~ 864 MHz / Tx : 824 ~ 849 MHz
 LTE Band 41 : 2555MHz~2655MHz
 LTE Band 1 : Rx : 2110 ~ 2170 MHz / Tx : 1920 ~ 1980 MHz
 LTE Band 3 : Rx : 1805 ~ 1880 MHz / Tx : 1710 ~ 1785 MHz
 LTE Band 7 : Rx : 2620 ~ 2690 MHz / Tx : 2500 ~ 2570 MHz
 LTE Band 8 : Rx : 925 ~ 960 MHz / Tx : 880 ~ 915 MHz
 LTE B38: 2570~2620 MHz
 LTE B39: 1880~1920 MHz
 LTE B40: 2300~2400 MHz
 TDSCDMA B34: 2010 ~ 2025 MHz
 TDSCDMA B39: 1880 ~ 1920 MHz
 BT/BLE/ANT+ : 2402 ~ 2480 MHz
 WLAN :
 2402 ~ 2480 MHz
 5180 ~ 5240 MHz / 5260 ~ 5320 MHz / 5500 ~ 5700 MHz
 5745 ~ 5825 MHz
 DTS : 2412 ~ 2472 MHz
 UNII : Ch144,142,138 not support



8. Level 3 Repair

8-3. Flow Chart of Troubleshooting



Oscilloscope



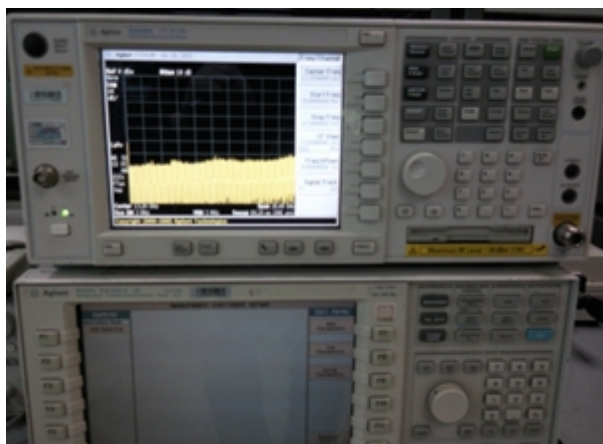
Digital Multimeter



Power Supply



+ driver, ESD Safe Tweezer



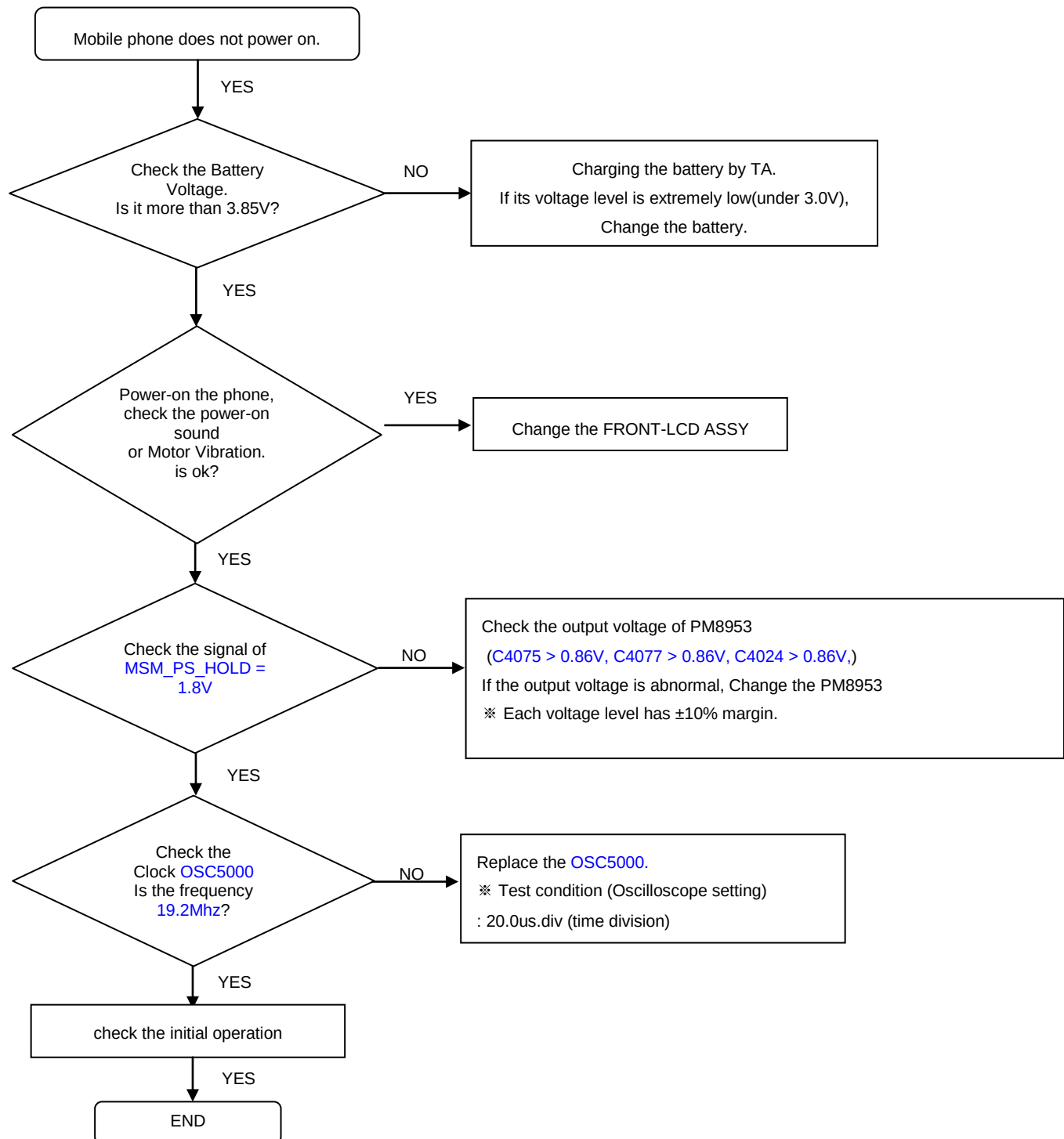
8960 & Spectrum Analyzer



Soldering iron

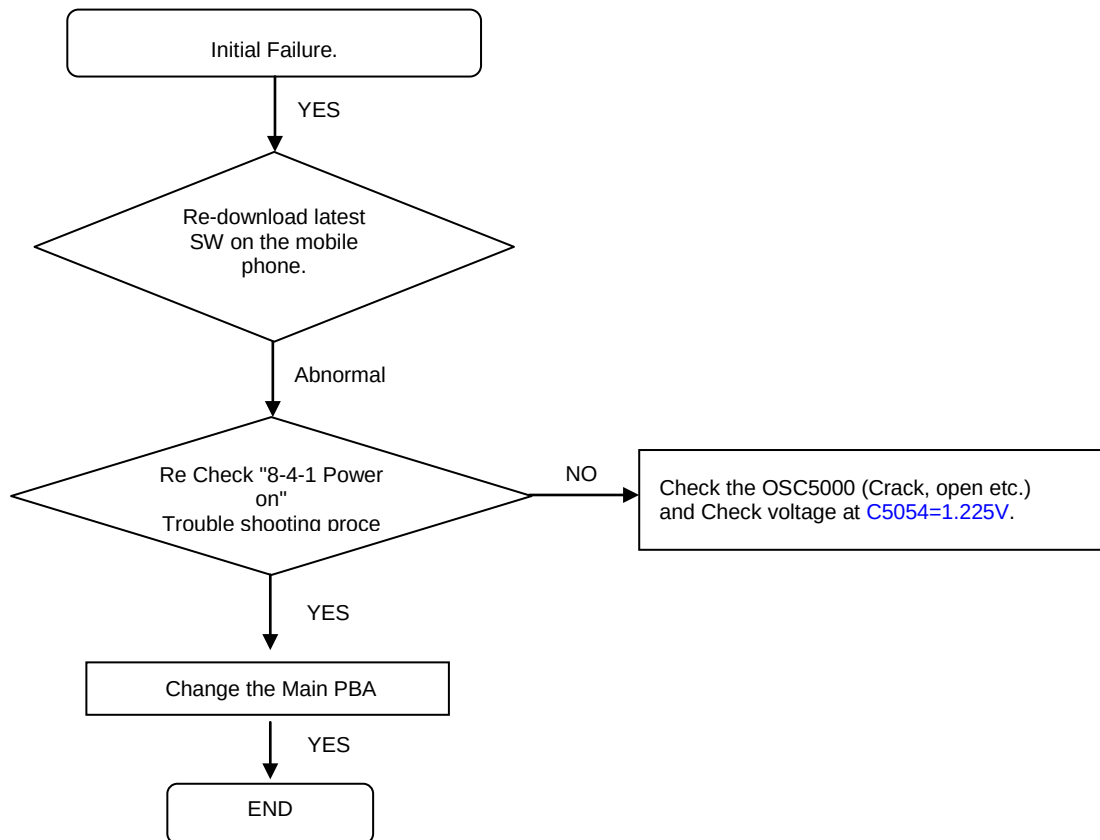
8. Level 3 Repair

8-4-1. Power On



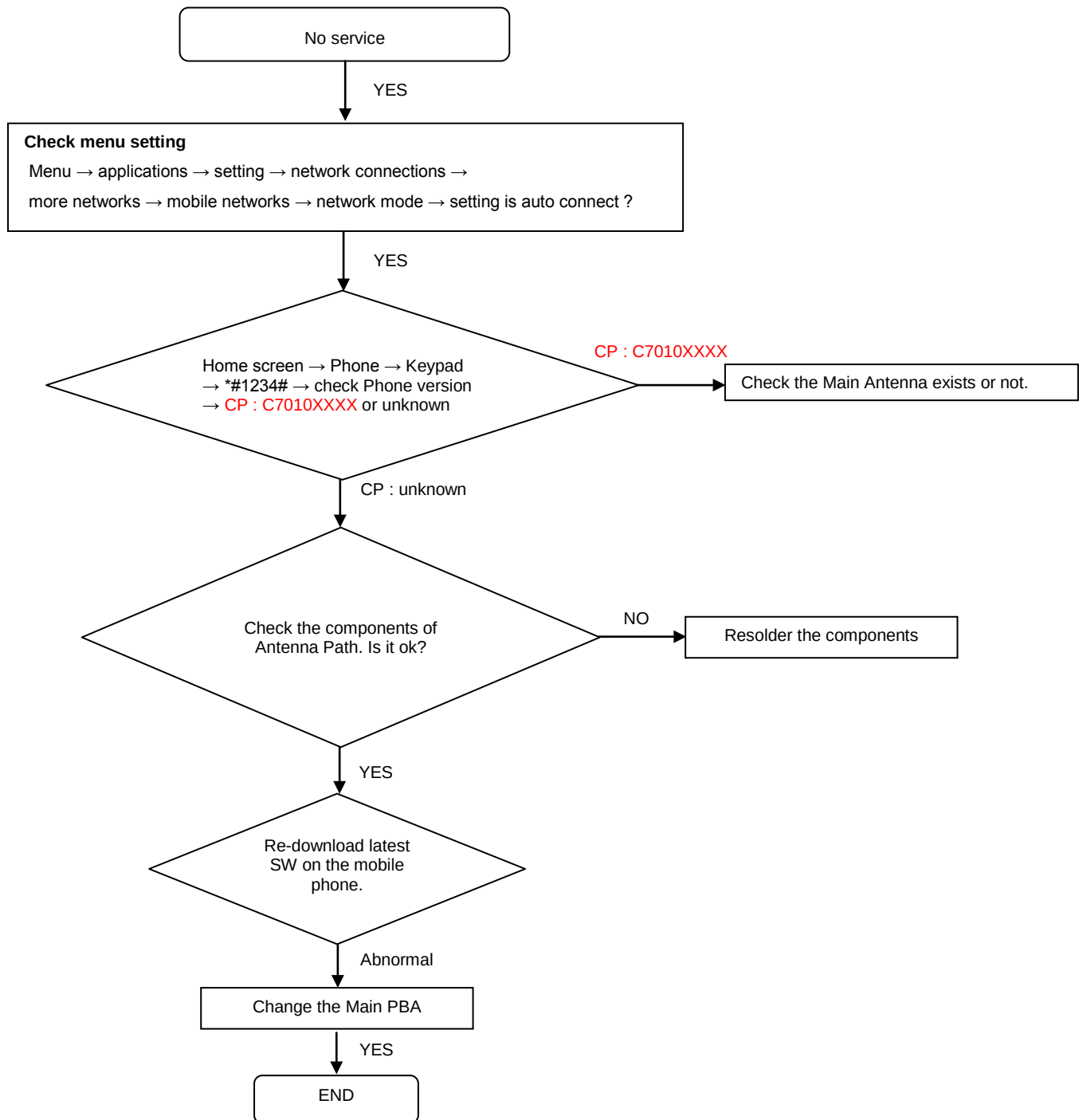
8. Level 3 Repair

8-4-2. Initial



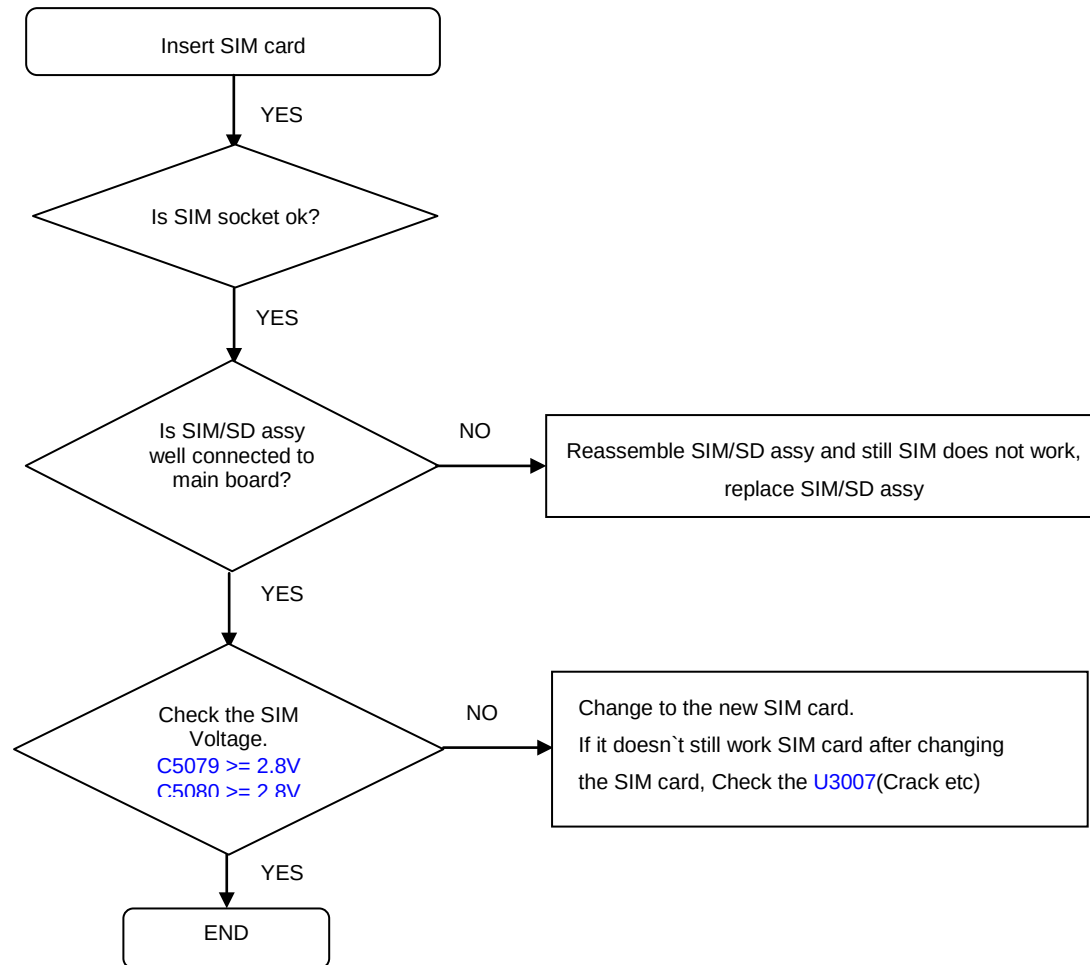
8. Level 3 Repair

8-4-3. No Service



8. Level 3 Repair

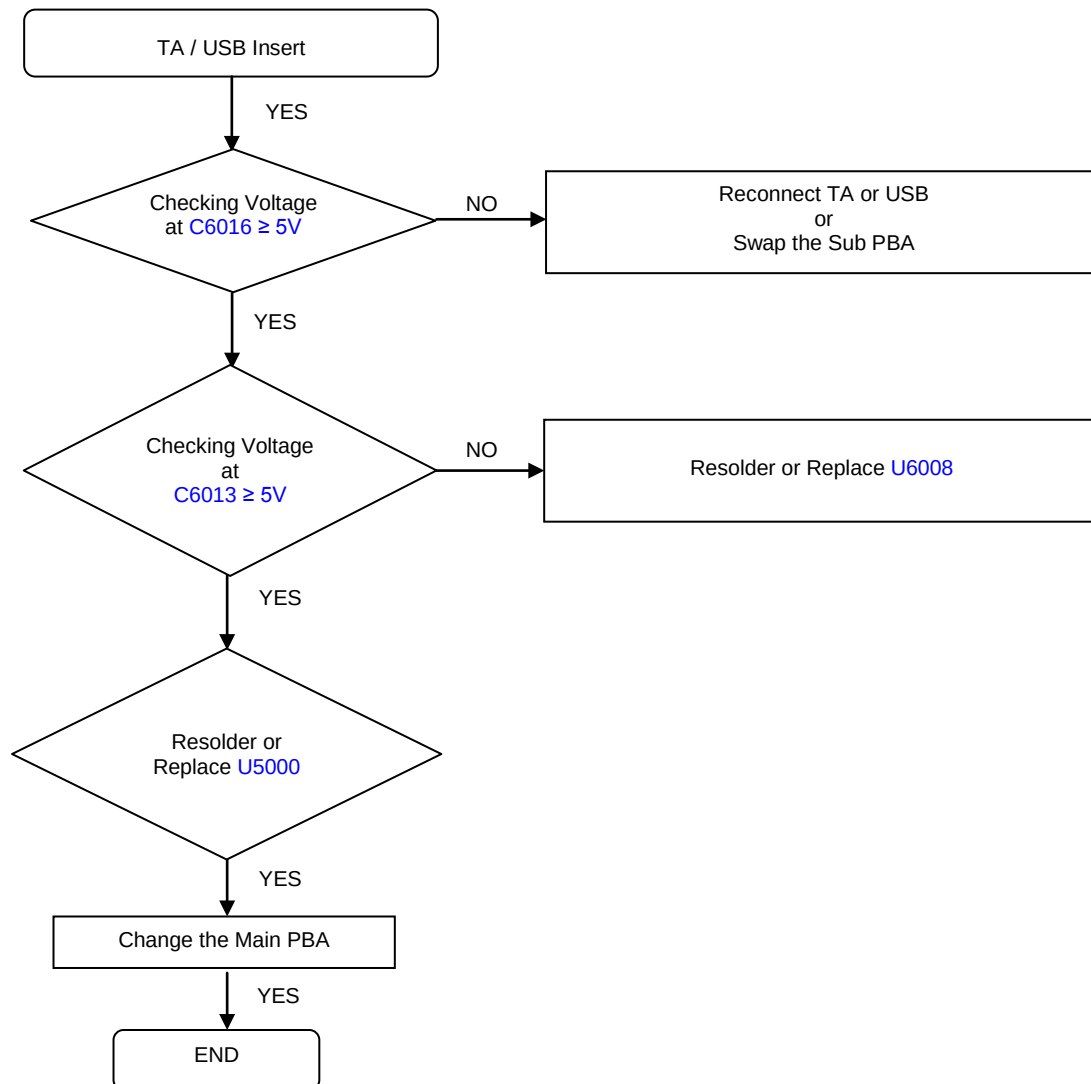
8-4-4. Sim Part



8. Level 3 Repair

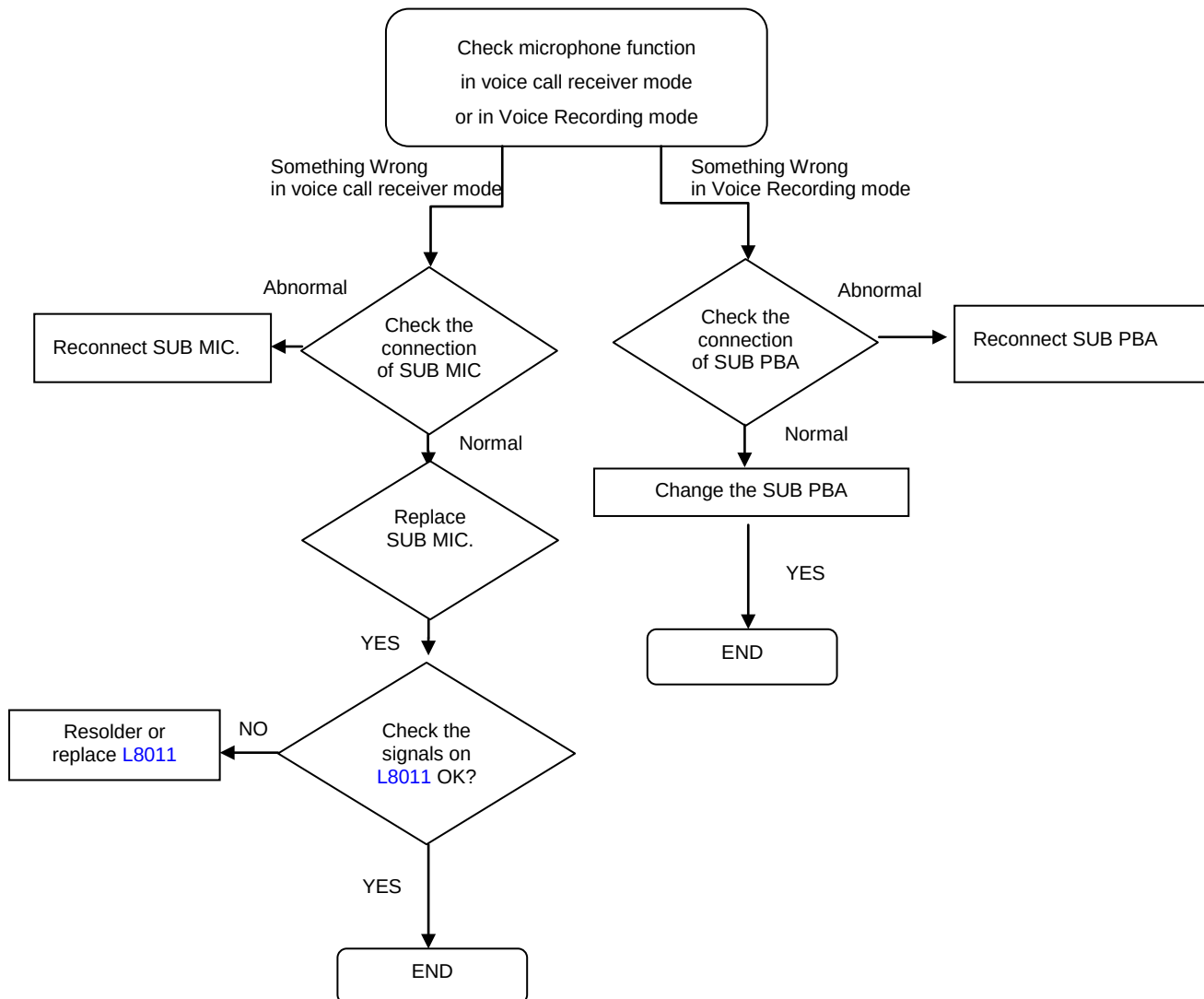
8-4-5. Charging Part

- First of all, Please check the battery status.



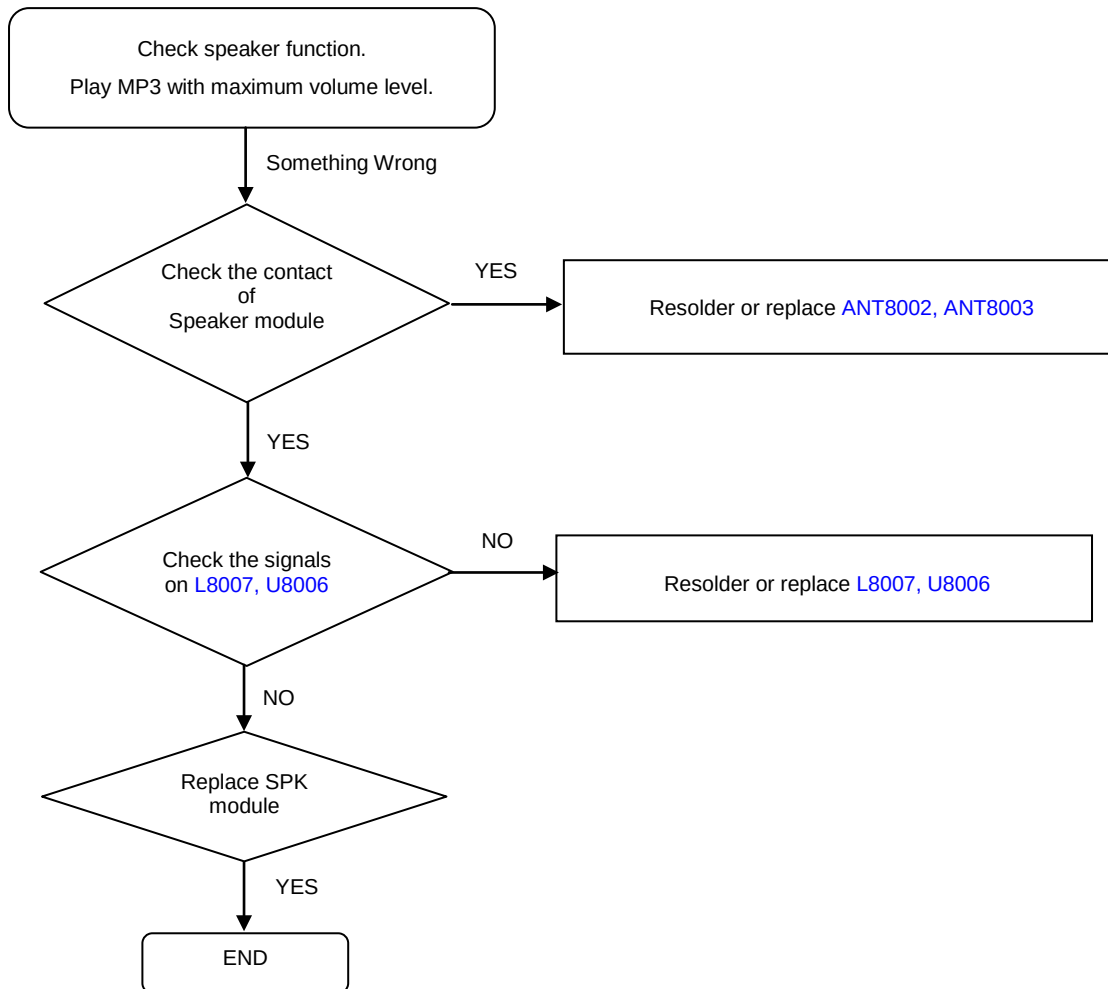
8. Level 3 Repair

8-4-6. Microphone Part



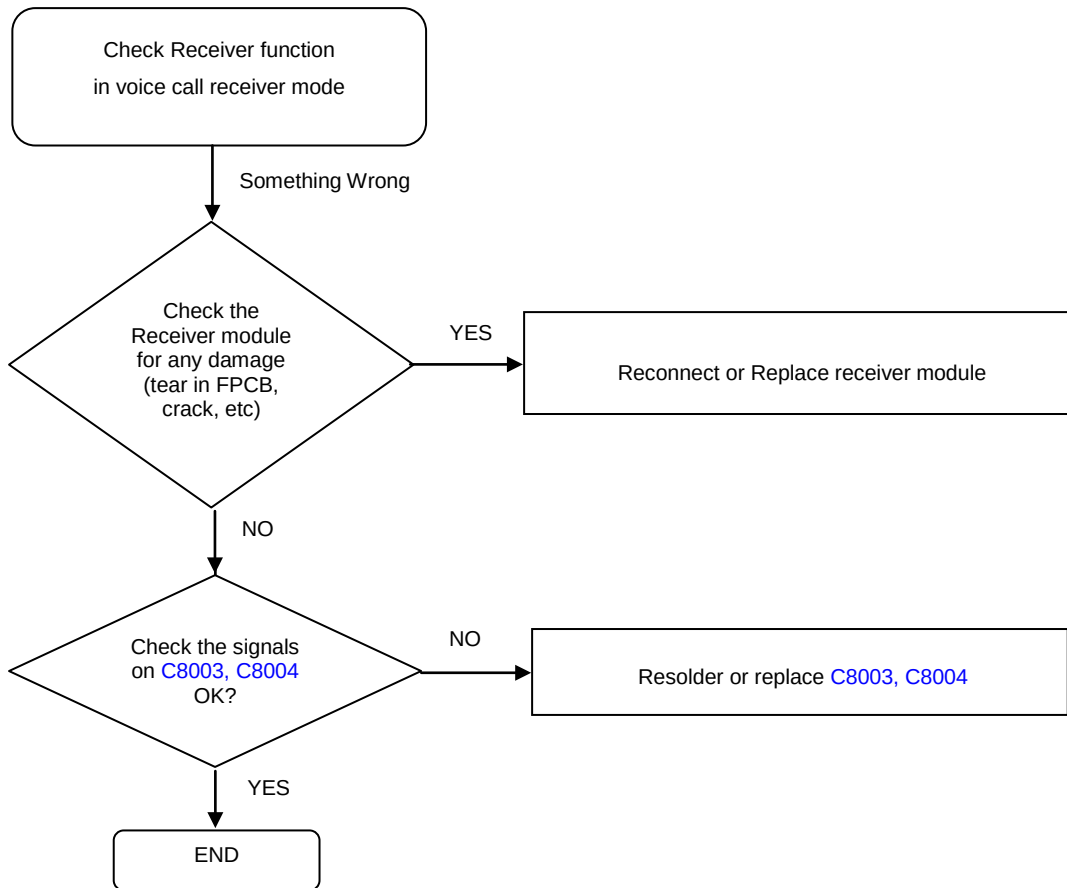
8. Level 3 Repair

8-4-7. Speaker Part



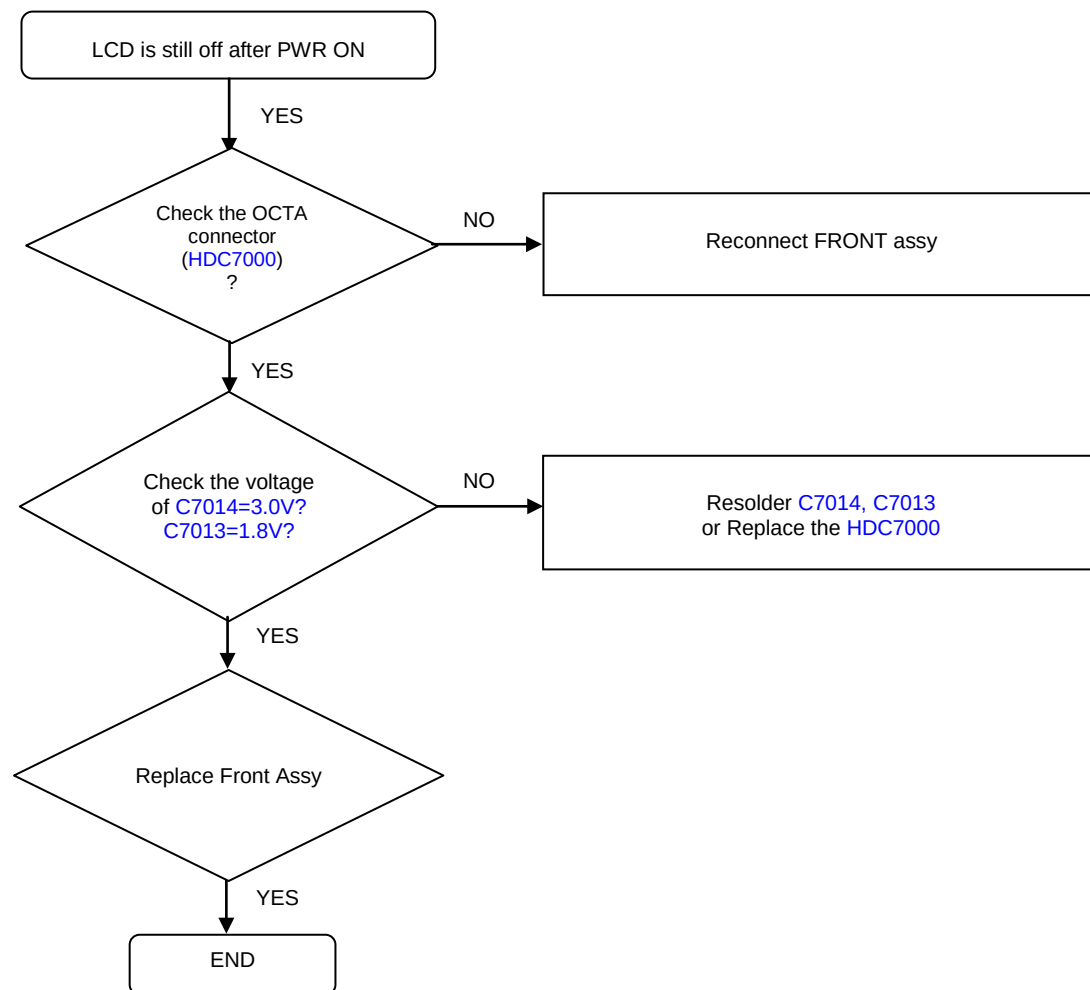
8. Level 3 Repair

8-4-8. Receiver Part



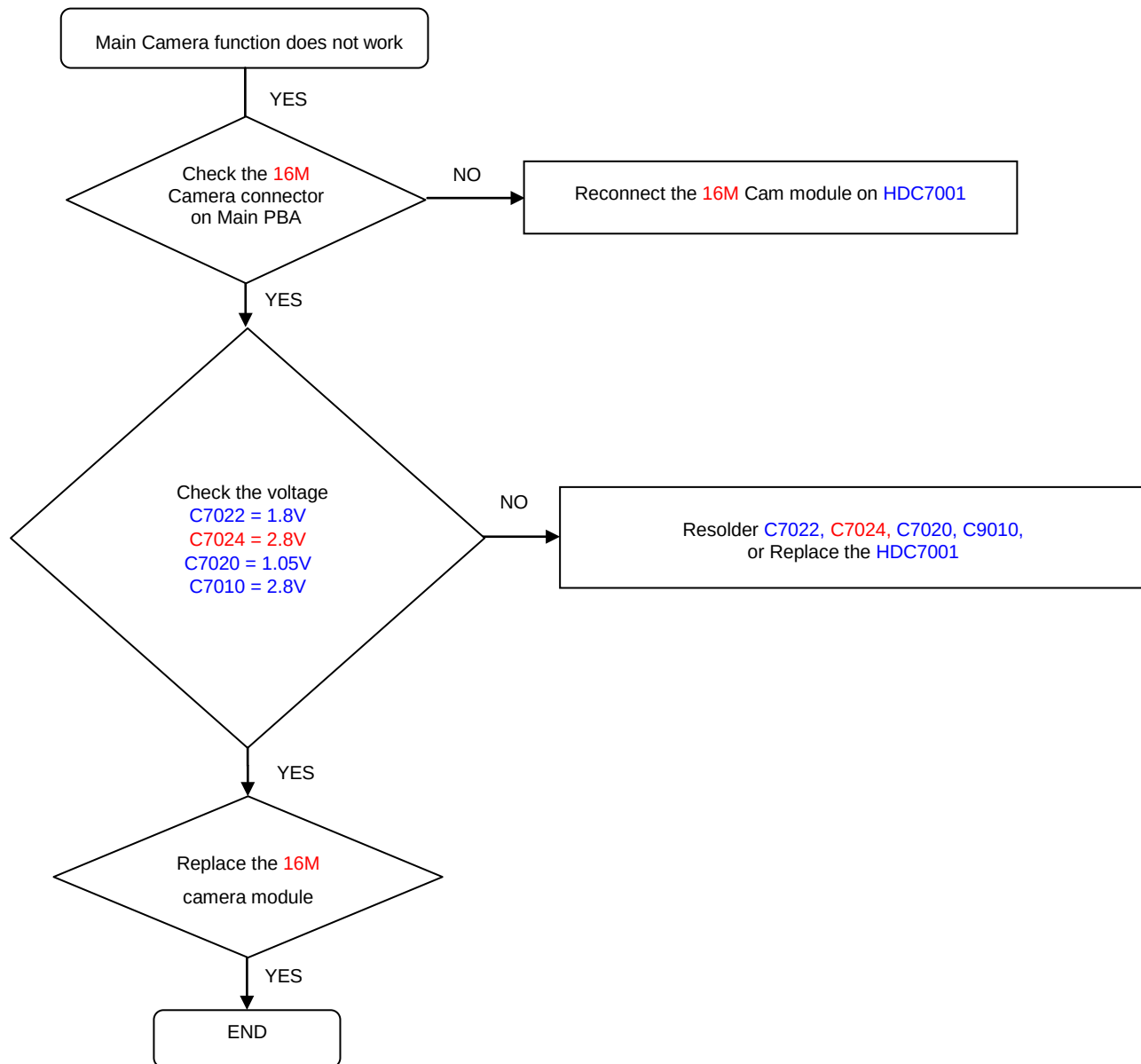
8. Level 3 Repair

8-4-9. OCTA



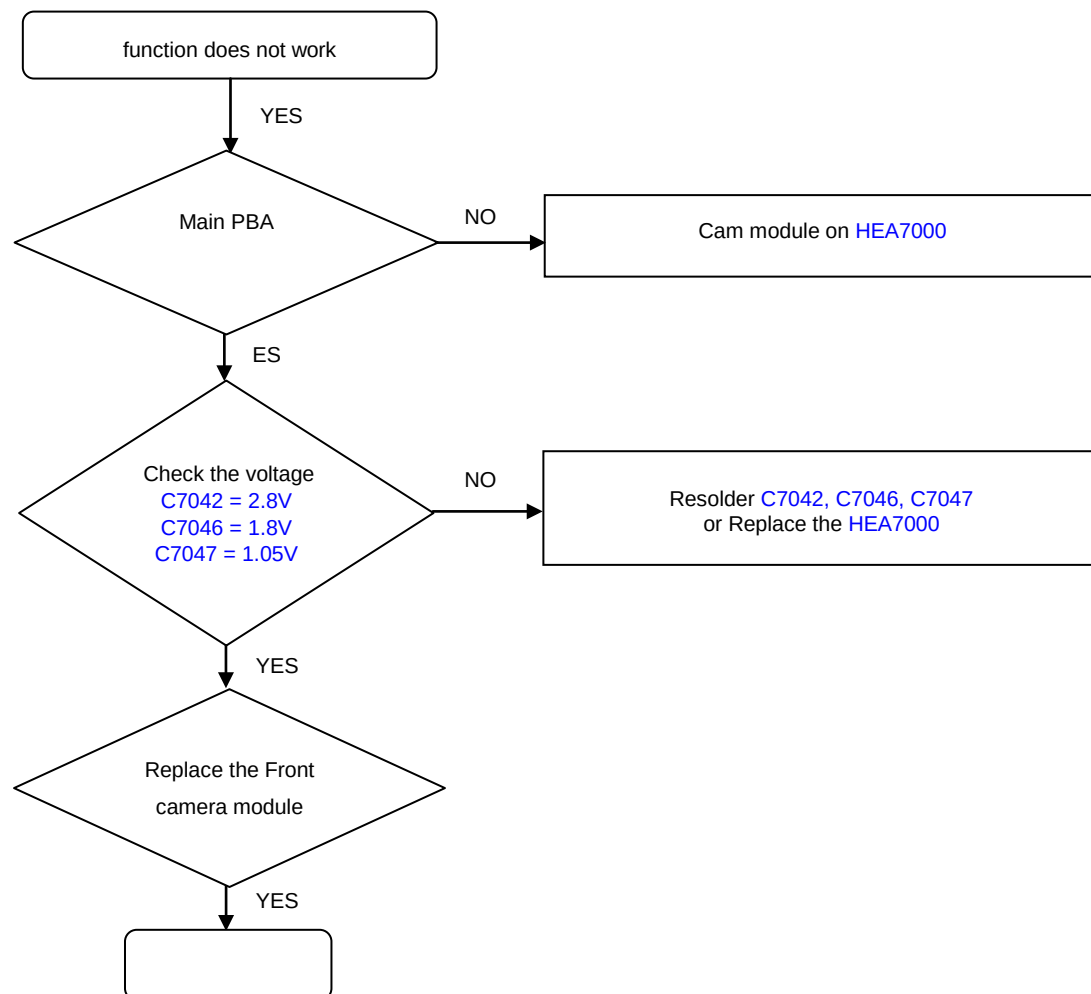
8. Level 3 Repair

8-4-10. 16M CAMERA



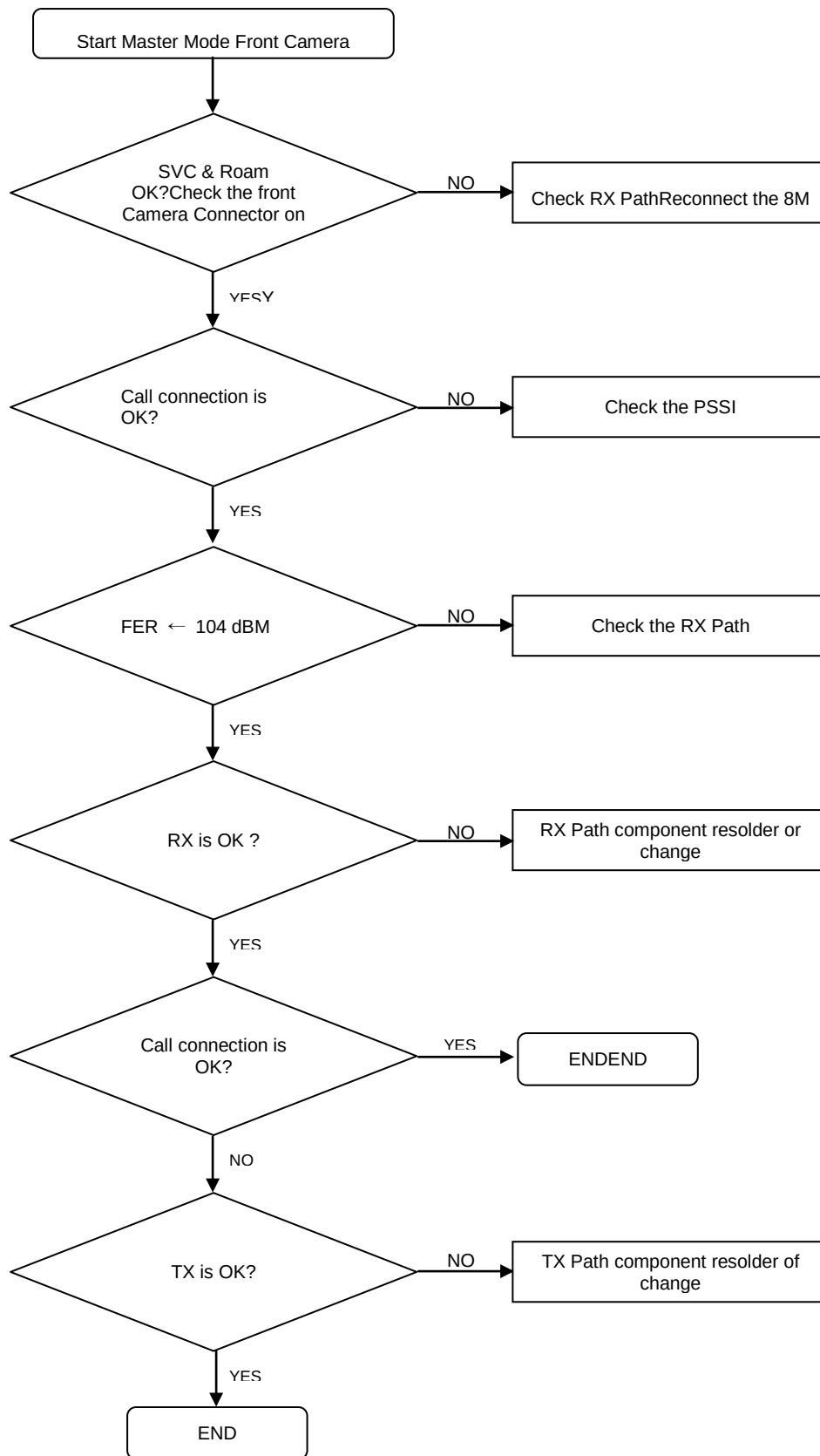
8. Level 3 Repair

8-4-11. 16M CAMERA



8. Level 3 Repair

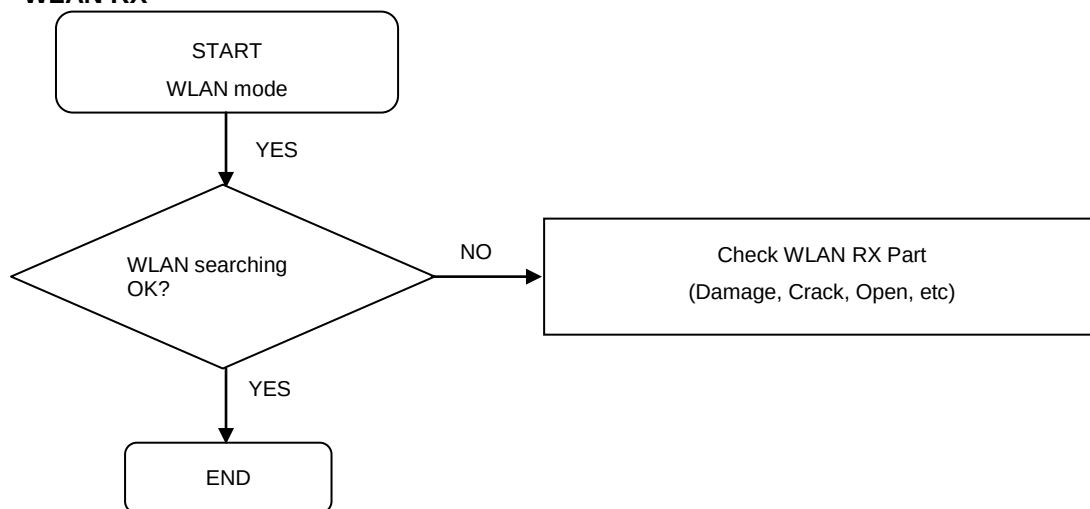
8-4-12. RF Section(LTE,WCDMA,TDSCDMA,CDMA,GSM)



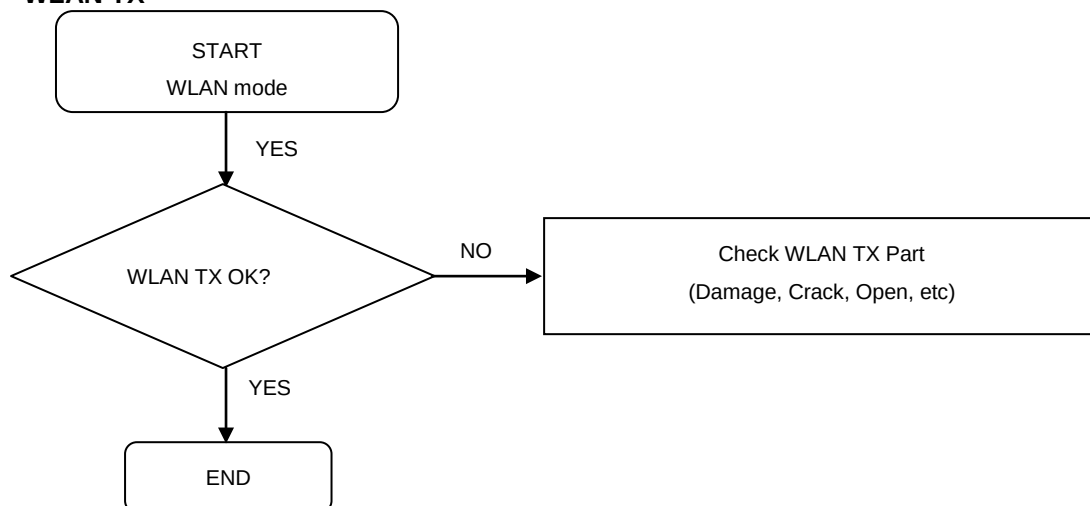
8. Level 3 Repair

8-4-13. RF Section(WLAN / BT)

- WLAN RX

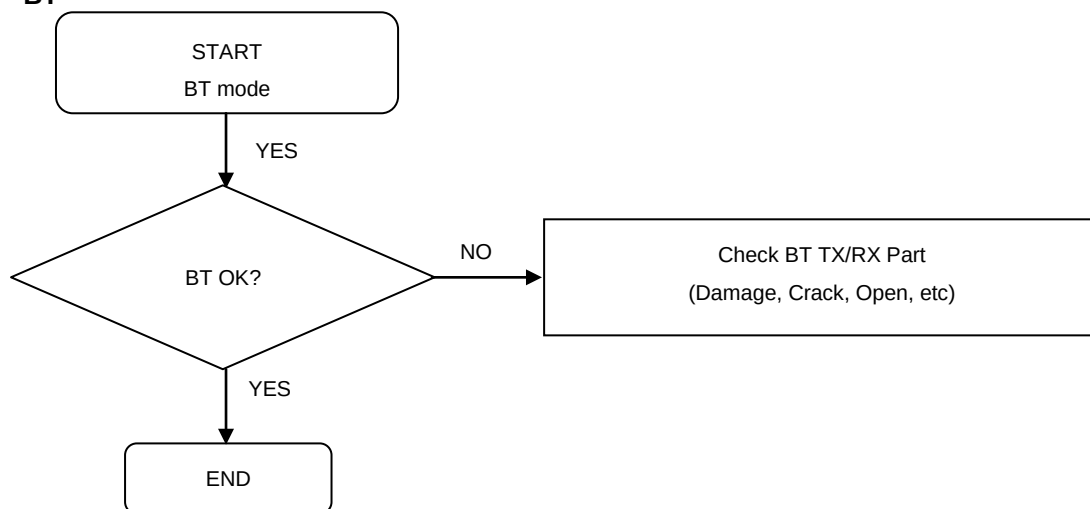


- WLAN TX

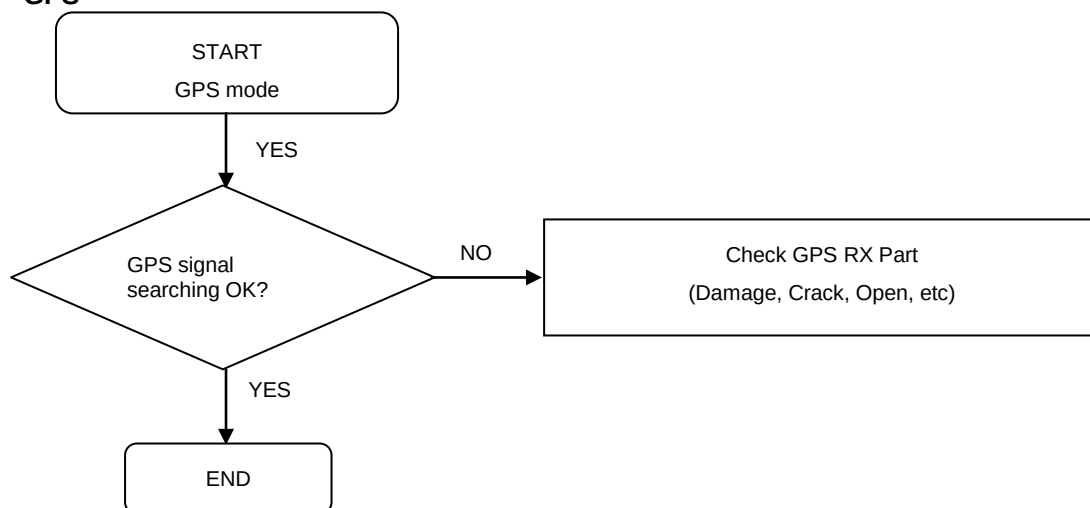


8. Level 3 Repair

- BT

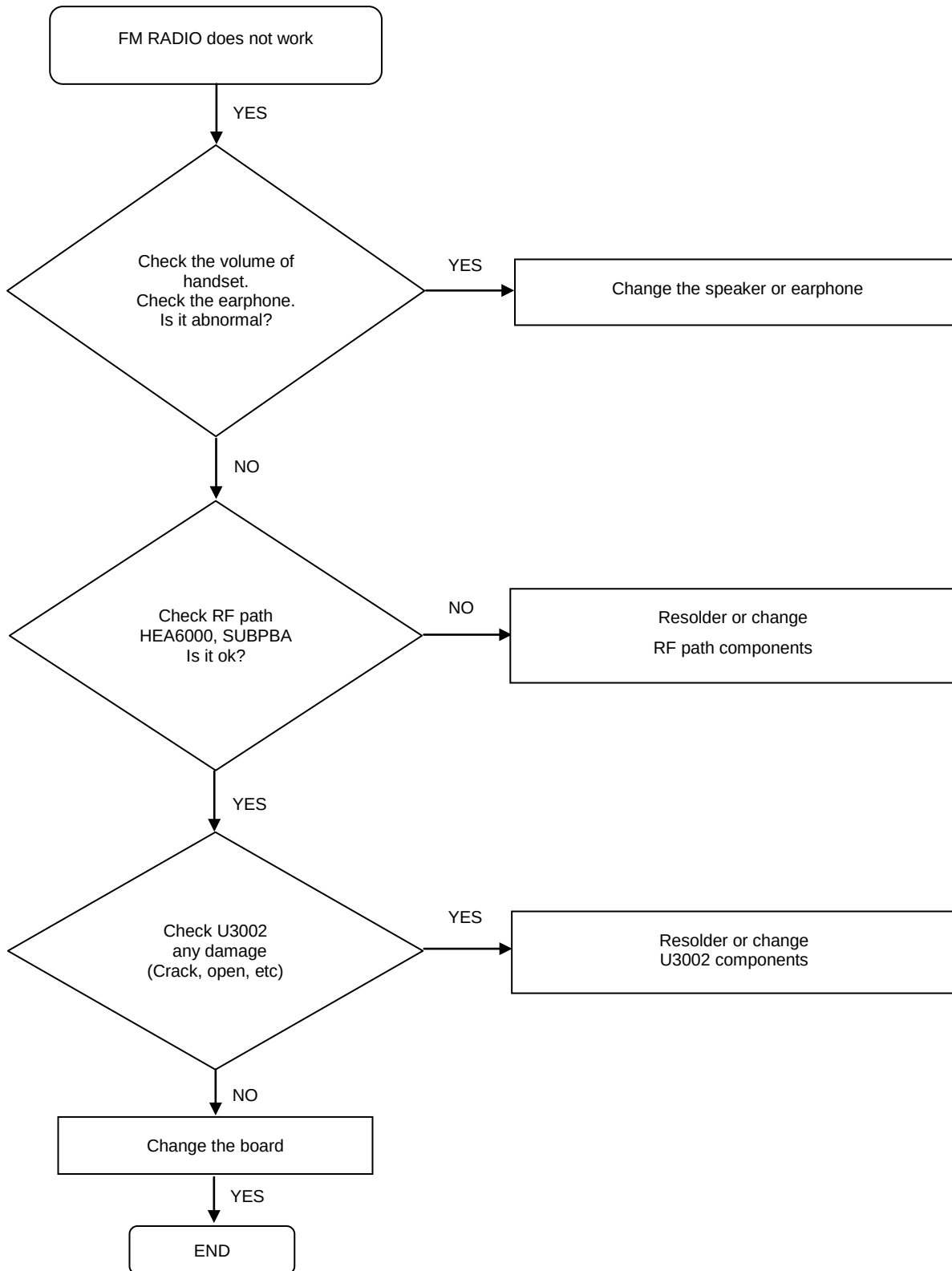


- GPS

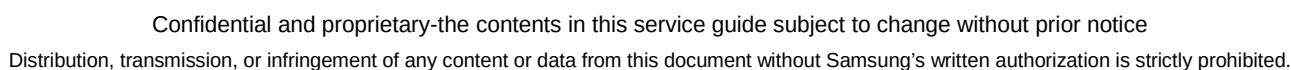


8. Level 3 Repair

8-4-14. FM RADIO



8-4-15. NFC



UCP400(AP)

[illegible]

5. MAIN Electrical Parts List

Design Loc	Parts Code	Description
0403-001972	ZD5000	DIODE-ZENER
0404-001250	D5001	DIODE-SCHOTTKY
0404-001782	D3000,D6001	DIODE-SCHOTTKY
0406-001459	ZD5001,ZD5002,ZD5014	DIODE-TVS
0406-001459	ZD5015	DIODE-TVS
0406-001465	ZD6000,ZD6001	DIODE-TVS
0406-001505	ZD5007,ZD5016,ZD5017	DIODE-TVS
0406-001505	ZD5018,ZD5019,ZD5020	DIODE-TVS
0406-001505	ZD5028	DIODE-TVS
0406-001542	ZD5006	DIODE-TVS
0406-001592	ZD5025,ZD5027,ZD6006	DIODE-TVS
0406-001592	ZD6007,ZD7000,ZD7001	DIODE-TVS
0406-001592	ZD7002,ZD7003,ZD7004	DIODE-TVS
0406-001592	ZD7005,ZD7006,ZD8000	DIODE-TVS
0406-001592	ZD8001	DIODE-TVS
0406-001682	ZD6008,ZD6009	DIODE-TVS
0406-001709	ZD6010,ZD6011,ZD6012	DIODE-TVS
0406-001709	ZD7009	DIODE-TVS
0406-001717	ZD5003,ZD5004,ZD5012	DIODE-TVS
0406-001717	ZD5013,ZD5021,ZD5022	DIODE-TVS
0406-001717	ZD5023,ZD5024,ZD5026	DIODE-TVS
0406-001717	ZD6002,ZD6003,ZD6013	DIODE-TVS
0406-001717	ZD6014	DIODE-TVS
0406-001728	D5000,ZD6015,ZD6016	DIODE-TVS
0505-002088	Q5001,TR6000,TR7000	FET-SILICON
0505-002388	Q5000	FET-SILICON
0601-001905	LED5001	LED
0601-003575	LED5000	LED
0903-002171	U7010	IC-MICROCONTROLLER
1001-001655	U5001	IC-ANALOG SWITCH
1001-001911	U1001	IC-RF SWITCH
1001-001913	U1006	IC-RF SWITCH
1001-001992	U1008	IC-RF SWITCH
1001-001999	U2001	IC-RF SWITCH
1001-002040	U1007	IC-RF SWITCH
1003-002415	U7011	IC-MOTOR DRIVER
1108-000624	UME4000	IC-MCP
1201-003717	U1015	IC-IF/RF AMP

5. MAIN Electrical Parts List

1201-003719	U1013,U1014	IC-IF/RF AMP
1201-003777	U1009	IC-IF/RF AMP
1201-003786	U3005	IC-GPS AMP
1201-003964	U8000,U8002	IC-AUDIO AMP
1201-004091	PAM1000	IC-POWER AMP
1203-006817	U7002,U7005	IC-POSI.FIXED REG.
1203-008249	U6010,U7007	IC-POSI.FIXED REG.
1203-008250	U3003,U3004,U7004	IC-POSI.FIXED REG.
1203-008250	U7013	IC-POSI.FIXED REG.
1203-008251	U7000,U7012,U8001	IC-POSI.FIXED REG.
1203-008326	U7001	IC-POSI.FIXED REG.
1203-008379	U2000	IC-POWER SUPERVISOR
1203-008395	U6006	IC-POSI.FIXED REG.
1203-008603	U5005	IC-MULTI REG.
1203-008700	U5004	IC-BATTERY
1203-008717	U7003	IC-DC/DC CONVERTER
1203-008791	U5002	IC-POWER SUPERVISOR
1203-008819	U5000	IC-POWER SUPERVISOR
1203-008830	U6008	IC-VOL. DETECTOR
1205-005037	U6002	IC-COMM. CONTROLLER
1205-005106	U5006	IC-SWITCH
1205-005142	U3002	IC-WIFI
1205-005228	U1012	IC-TRANSCEIVER
1205-005652	U6009	IC-POWER DRIVER
1205-005680	U3007	IC-NFC/RFID
1205-005714	UCP4000	IC-MODEM
1205-005750	U8003	IC-CODEC
1209-002344	U7006	IC-ACCEL./GYRO SENSOR
1209-002401	U7009	IC-MAGNETIC SENSOR
1209-002474	U7008	IC-OPTICS SENSOR
1404-001664	TH4000,TH5000	THERMISTOR-NTC
1404-001724	TH2000	THERMISTOR-NTC
1405-001395	C3041,V3000,V5000	VARISTOR
1405-001395	V5001,VR2000,VR2001	VARISTOR
1405-001395	VR3000,VR3002,VR7000	VARISTOR
1405-001395	VR7001	VARISTOR
1405-001415	V1000,V1001,V3001	VARISTOR
1405-001415	VR3001	VARISTOR
2007-001284	R6027,R6029	R-CHIP

5. MAIN Electrical Parts List

2007-003015	R6035,R6036	R-CHIP
2007-007107	R2000	R-CHIP
2007-007136	R3005	R-CHIP
2007-007741	R4033,R4037,R4041	R-CHIP
2007-007741	R4043,R5025,R5026	R-CHIP
2007-007741	R5027,R5035,R5042	R-CHIP
2007-007741	R5048,R6026,R6028	R-CHIP
2007-007741	R6038,R6039,R7002	R-CHIP
2007-007741	R7008	R-CHIP
2007-007798	R5057	R-CHIP
2007-008210	R5036	R-CHIP
2007-008420	R4002,R4030,R5018	R-CHIP
2007-008486	R5010,R5023	R-CHIP
2007-008588	R3006,R3007,R7015	R-CHIP
2007-008588	R7016,R7017,R7018	R-CHIP
2007-008766	R4004	R-CHIP
2007-008774	R5052,R5053,R5054	R-CHIP
2007-009111	R7009	R-CHIP
2007-009155	R5004,R5013	R-CHIP
2007-009157	R3001,R4032,R5012	R-CHIP
2007-009157	R5014,R5015,R5016	R-CHIP
2007-009157	R6017,R6018,R6019	R-CHIP
2007-009157	R6046,R7011,R7020	R-CHIP
2007-009158	R5003	R-CHIP
2007-009182	R7019	R-CHIP
2007-009212	R3000,R6000,R6005	R-CHIP
2007-009212	R6009,R6025,R8008	R-CHIP
2007-009315	R5031,R5032,R5033	R-CHIP
2007-009315	R5037,R5038,R5039	R-CHIP
2007-009315	R5040,R5041,R5043	R-CHIP
2007-009315	R5044,R5046	R-CHIP
2007-009352	R4018,R4021,R6003	R-CHIP
2007-009352	R7013,R7014	R-CHIP
2007-009408	R4010,R4011,R4012	R-CHIP
2007-009408	R4013,R4014,R4015	R-CHIP
2007-009408	R4016,R4017,R4019	R-CHIP
2007-009408	R4020,R4022,R4023	R-CHIP
2007-009408	R4024,R4025,R4028	R-CHIP
2007-009408	R4029,R4051,R4052	R-CHIP

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2007-009408	R6015,R6016	R-CHIP
2007-009410	R1003	R-CHIP
GH80-03321A	R4049	R-CHIP
2007-009766	R5000	R-CHIP
2007-009801	R7000	R-CHIP
2007-009805	R1000,R3002	R-CHIP
2007-009920	R4008,R4046,R4048	R-CHIP
2007-009969	R8009	R-CHIP
2007-010029	R6022,R6023	R-CHIP
2007-010202	R5045,R6020	R-CHIP
2007-010233	R4031	R-CHIP
2007-010685	R8018,R8019,R8020	R-CHIP
2007-010856	R5034	R-CHIP
2007-011648	R8041	R-CHIP
2007-011763	R4026	R-CHIP
2203-000425	C6026,C6027	C-CER,CHIP
2203-000438	C3066	C-CER,CHIP
2203-000725	C7067	C-CER,CHIP
2203-000812	C3067,C3077	C-CER,CHIP
2203-000854	C6016	C-CER,CHIP
2203-001072	C3070,C3071	C-CER,CHIP
2203-001153	C5062,C6037,C6038	C-CER,CHIP
2203-001239	C3068,C3075	C-CER,CHIP
2203-005344	C5030	C-CER,CHIP
2203-005682	C1000,C1029,C1050	C-CER,CHIP
2203-005682	C1062,C1064,C1078	C-CER,CHIP
2203-005682	C1083,C1102,C1127	C-CER,CHIP
2203-005682	C1131,C1136,C1137	C-CER,CHIP
2203-005682	C1143,C2007,C2013	C-CER,CHIP
2203-005682	C2016,C2017,C2019	C-CER,CHIP
2203-005682	C2023,C3050,C3054	C-CER,CHIP
2203-005682	C5071,C5081,C7019	C-CER,CHIP
2203-005682	C7021,C7023,C7025	C-CER,CHIP
2203-005682	C7076,C7077,C8046	C-CER,CHIP
2203-005682	L1012,L1045	C-CER,CHIP
2203-005717	C2010	C-CER,CHIP
2203-005719	C1072	C-CER,CHIP
2203-005725	C1026,C1042,C1065	C-CER,CHIP
2203-005725	C1079,C4086	C-CER,CHIP

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2203-005727	C3039	C-CER,CHIP
2203-005729	C1005,C5000	C-CER,CHIP
2203-005731	C3000,C3006,C3032	C-CER,CHIP
2203-005731	C3036,C6033	C-CER,CHIP
2203-005732	C6034,C8003,C8004	C-CER,CHIP
2203-005732	C8022,C8036,C8045	C-CER,CHIP
2203-005732	C8048	C-CER,CHIP
2203-005734	C7074	C-CER,CHIP
2203-005736	C1004,C1012,C1033	C-CER,CHIP
2203-005736	C1034,C1047,C1055	C-CER,CHIP
2203-005736	C1058,C1073,C1082	C-CER,CHIP
2203-005736	C1084,C1098,C1101	C-CER,CHIP
2203-005736	C1109,C1110,C1113	C-CER,CHIP
2203-005736	C1139,C1140,C1148	C-CER,CHIP
2203-005736	C2006,C3009,C3016	C-CER,CHIP
2203-005736	C3018,C3019,C3021	C-CER,CHIP
2203-005736	C3022,C3030,C3055	C-CER,CHIP
2203-005736	C3061,C5076,C7005	C-CER,CHIP
2203-005736	C7006,L1004	C-CER,CHIP
2203-005777	C1054,C1070,C1089	C-CER,CHIP
2203-005777	C3013,C3029,L1009	C-CER,CHIP
2203-005779	C3040	C-CER,CHIP
2203-005789	C1028,C1060,C1069	C-CER,CHIP
2203-005789	C1074,C1085,L1002	C-CER,CHIP
2203-005792	C1090,C1138,C1145	C-CER,CHIP
2203-005792	C3058	C-CER,CHIP
2203-005806	C4087,C5050	C-CER,CHIP
2203-006048	C7064,C7065	C-CER,CHIP
2203-006121	C8047	C-CER,CHIP
2203-006187	C3005	C-CER,CHIP
2203-006194	C3010,C3011,C3012	C-CER,CHIP
2203-006194	C3017,C3020,C3023	C-CER,CHIP
2203-006318	C1018,C1053	C-CER,CHIP
2203-006348	C6013	C-CER,CHIP
2203-006399	C4088,C4092,C4095	C-CER,CHIP
2203-006399	C4096,C5003,C5013	C-CER,CHIP
2203-006399	C5067,C5068,C5082	C-CER,CHIP
2203-006399	C6000,C6001,C6035	C-CER,CHIP
2203-006399	C7001,C7013,C7014	C-CER,CHIP

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2203-006399	C7022,C7024,C7028	C-CER,CHIP
2203-006399	C7039,C7069	C-CER,CHIP
2203-006400	C1115,C3001	C-CER,CHIP
2203-006401	C6020,C6021	C-CER,CHIP
2203-006410	C1045,C1114,C7016	C-CER,CHIP
2203-006410	C7017	C-CER,CHIP
2203-006423	C1014,C1067,C1092	C-CER,CHIP
2203-006423	C1093,C1097,C1104	C-CER,CHIP
2203-006423	C1105,C1106,C1107	C-CER,CHIP
2203-006423	C1108,C1112,C1118	C-CER,CHIP
2203-006423	C1121,C1135,C3065	C-CER,CHIP
2203-006423	C4000,C4001,C4004	C-CER,CHIP
2203-006423	C4005,C4008,C4012	C-CER,CHIP
2203-006423	C4014,C4015,C4018	C-CER,CHIP
2203-006423	C4019,C4026,C4027	C-CER,CHIP
2203-006423	C4029,C4031,C4032	C-CER,CHIP
2203-006423	C4033,C4035,C4036	C-CER,CHIP
2203-006423	C4037,C4038,C4039	C-CER,CHIP
2203-006423	C4040,C4041,C4042	C-CER,CHIP
2203-006423	C4043,C4044,C4045	C-CER,CHIP
2203-006423	C4046,C4047,C4048	C-CER,CHIP
2203-006423	C4049,C4050,C4051	C-CER,CHIP
2203-006423	C4052,C4053,C4054	C-CER,CHIP
2203-006423	C4057,C4059,C4061	C-CER,CHIP
2203-006423	C4062,C4063,C4064	C-CER,CHIP
2203-006423	C4065,C4066,C4068	C-CER,CHIP
2203-006423	C4069,C4070,C4079	C-CER,CHIP
2203-006423	C4080,C4082,C4083	C-CER,CHIP
2203-006423	C4094,C4097,C4099	C-CER,CHIP
2203-006423	C4100,C4103,C4104	C-CER,CHIP
2203-006423	C4105,C5022,C5061	C-CER,CHIP
2203-006423	C5079,C5080,C7045	C-CER,CHIP
2203-006423	C7051,C7058,C7059	C-CER,CHIP
2203-006423	C7060,C7061,C7062	C-CER,CHIP
2203-006423	C8009,C8011,C8013	C-CER,CHIP
2203-006562	C5036,C5066,C6011	C-CER,CHIP
2203-006562	C6023,C6036,C7000	C-CER,CHIP
2203-006562	C7015,C7027,C7029	C-CER,CHIP
2203-006562	C7040,C7049,C7070	C-CER,CHIP

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2203-006562	C7072	C-CER,CHIP
2203-006604	C1063,C5072	C-CER,CHIP
2203-006611	C1059,C3008,L3007	C-CER,CHIP
2203-006648	C3027,C4017,C6039	C-CER,CHIP
2203-006665	C1056,C3004	C-CER,CHIP
2203-006674	C1036,C1081	C-CER,CHIP
2203-006707	C1117,C1120,C1147	C-CER,CHIP
2203-006839	C1008,C1015,C1023	C-CER,CHIP
2203-006839	C1116,C3033,C4106	C-CER,CHIP
2203-006839	C4107,C5027,C5037	C-CER,CHIP
2203-006839	C5060,C7066,C8015	C-CER,CHIP
2203-006841	C6022	C-CER,CHIP
2203-006872	C7030,C7041	C-CER,CHIP
2203-007143	C6008	C-CER,CHIP
2203-007194	C7026,C7050	C-CER,CHIP
2203-007240	C4024,C4025,C4071	C-CER,CHIP
2203-007240	C4075,C4077,C4078	C-CER,CHIP
2203-007269	C7032,C7034	C-CER,CHIP
2203-007270	C7007	C-CER,CHIP
2203-007271	C5045,C5053,C5056	C-CER,CHIP
2203-007271	C7012	C-CER,CHIP
2203-007317	C2000,C3028,C4093	C-CER,CHIP
2203-007317	C4102,C5019,C5023	C-CER,CHIP
2203-007317	C5025,C7044,C7054	C-CER,CHIP
2203-007317	C7063,C8028,C8030	C-CER,CHIP
2203-007317	C8033,C8035,C8037	C-CER,CHIP
2203-007317	C8038,C8039,C8042	C-CER,CHIP
2203-007317	C8043	C-CER,CHIP
2203-007391	C1091,C2004,C2021	C-CER,CHIP
2203-007391	C3024,C3025,C4090	C-CER,CHIP
2203-007391	C4101	C-CER,CHIP
2203-007393	C6018	C-CER,CHIP
2203-007449	C1030,C1099,C4006	C-CER,CHIP
2203-007449	C4007,C4013,C4020	C-CER,CHIP
2203-007449	C4028,C4030,C4055	C-CER,CHIP
2203-007449	C4060,C4067,C4073	C-CER,CHIP
2203-007449	C4084,C4089,C4091	C-CER,CHIP
2203-007449	C4098,C5004,C5005	C-CER,CHIP
2203-007449	C5006,C5008,C5010	C-CER,CHIP

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2203-007449	C5011,C5012,C5014	C-CER,CHIP
2203-007449	C5020,C5024,C5026	C-CER,CHIP
2203-007449	C5031,C5049,C5070	C-CER,CHIP
2203-007449	C6012,C6028,C6029	C-CER,CHIP
2203-007449	C6030,C6031,C7038	C-CER,CHIP
2203-007449	C7046,C7071,C8024	C-CER,CHIP
2203-007449	C8025,C8029,C8031	C-CER,CHIP
2203-007449	C8032,C8034,C8044	C-CER,CHIP
2203-007456	C1007	C-CER,CHIP
2203-007474	C3003,C3015,C3026	C-CER,CHIP
2203-007474	C4058,C7057,C8010	C-CER,CHIP
2203-007474	C8012,C8014	C-CER,CHIP
2203-007634	C6014	C-CER,CHIP
2203-007693	C5001	C-CER,CHIP
2203-007775	C2002,C3031,C3034	C-CER,CHIP
2203-007775	C3035,C5034,C5039	C-CER,CHIP
2203-007775	C5040,C5041,C5043	C-CER,CHIP
2203-007775	C5044,C5046,C5047	C-CER,CHIP
2203-007781	C5028,C5029,C7031	C-CER,CHIP
2203-007781	C7033,C7037	C-CER,CHIP
2203-007795	C1094,C1095,C7009	C-CER,CHIP
2203-007795	C7010,C7020,C7042	C-CER,CHIP
2203-007795	C7043,C7047	C-CER,CHIP
2203-007796	C3042,C3043,C3045	C-CER,CHIP
2203-007796	C3047,C3048,C3049	C-CER,CHIP
2203-007796	C3063,C3064,C5058	C-CER,CHIP
2203-007796	C7011,C7052,C7068	C-CER,CHIP
2203-007796	C8005	C-CER,CHIP
2203-007854	C3069,C3073	C-CER,CHIP
2203-008123	C4034,C5054,C5055	C-CER,CHIP
2203-008242	C4056,C5073,C5074	C-CER,CHIP
2203-008242	C8017,C8023,C8027	C-CER,CHIP
2203-008243	C5059	C-CER,CHIP
2203-008572	C5057	C-CER,CHIP
2203-008860	C3052,C3053,C3057	C-CER,CHIP
2203-008860	C7008,C8008	C-CER,CHIP
2203-009328	C3002	C-CER,CHIP
2203-009551	C1009,C1016	C-CER,CHIP
2203-009618	C5064,C6017,C8006	C-CER,CHIP

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2203-009618	C8040	C-CER,CHIP
2203-009733	C1022,C2001,C3051	C-CER,CHIP
2203-009733	C5032,C5033,C5042	C-CER,CHIP
2203-009733	C5069,C5083,C6015	C-CER,CHIP
2203-009733	C6019,C8000,C8001	C-CER,CHIP
2203-009733	C8007,C8016,C8019	C-CER,CHIP
2203-009733	C8021,C8041	C-CER,CHIP
2203-009738	C4076	C-CER,CHIP
2203-009812	C5002	C-CER,CHIP
2703-002903	L2009	INDUCTOR-SMD
2703-002951	L1010	INDUCTOR-SMD
2703-002958	L1027,L1028,L1029	INDUCTOR-SMD
2703-002958	L1031,L3008	INDUCTOR-SMD
2703-002959	R7004,R7007	INDUCTOR-SMD
2703-002999	C1103	INDUCTOR-SMD
2703-003205	L7008,L7009	INDUCTOR-SMD
2703-003970	C1013,L1023	INDUCTOR-SMD
2703-004012	L1018	INDUCTOR-SMD
2703-004013	C2022,L1013,L1049	INDUCTOR-SMD
2703-004014	C1153,C3007,C3037	INDUCTOR-SMD
2703-004034	C2009,C3014,L2001	INDUCTOR-SMD
2703-004034	L2003,L2004,L2005	INDUCTOR-SMD
2703-004035	L1022	INDUCTOR-SMD
2703-004037	L1034	INDUCTOR-SMD
2703-004038	C1001,C2012,L1038	INDUCTOR-SMD
2703-004038	L1043,L8006	INDUCTOR-SMD
2703-004286	C1048,C1124,C1133	INDUCTOR-SMD
2703-004286	C3038,L1001,L1032	INDUCTOR-SMD
2703-004286	L1037	INDUCTOR-SMD
2703-004287	L3003	INDUCTOR-SMD
2703-004295	L3013,L6000,L6001	INDUCTOR-SMD
2703-004328	L1000	INDUCTOR-SMD
2703-004366	L1014,L1041	INDUCTOR-SMD
2703-004367	L1011,L1040	INDUCTOR-SMD
2703-004369	L1016,L1017,L1019	INDUCTOR-SMD
2703-004369	L1021,L1035	INDUCTOR-SMD
2703-004371	L5010	INDUCTOR-SMD
2703-004408	L1048,L3002	INDUCTOR-SMD
2703-004763	C1141,L1007	INDUCTOR-SMD

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2703-004764	C1040,C1044,C3056	INDUCTOR-SMD
2703-004862	L1025,L1026	INDUCTOR-SMD
2703-004911	C1027,L1015,L1046	INDUCTOR-SMD
2703-004911	L2007	INDUCTOR-SMD
2703-004976	L1003,L1006	INDUCTOR-SMD
2703-005058	C1041,L1033	INDUCTOR-SMD
2703-005066	L8000,L8005	INDUCTOR-SMD
2703-005073	L3014,L3015	INDUCTOR-SMD
2703-005085	C1051,L1030,L1036	INDUCTOR-SMD
2703-005085	L1044,L2006	INDUCTOR-SMD
2703-005087	L1008	INDUCTOR-SMD
2703-005100	L8007,U8006	INDUCTOR-SMD
2703-005101	L2000	INDUCTOR-SMD
2703-005117	L7001,L7003	INDUCTOR-SMD
2703-005118	L7002	INDUCTOR-SMD
2703-005136	L5001,L5009	INDUCTOR-SMD
2703-005219	L8001,L8002	INDUCTOR-SMD
2703-005339	L5004,L5005,L5008	INDUCTOR-SMD
2703-005366	C2005	INDUCTOR-SMD
2703-005505	L5002,L5003	INDUCTOR-SMD
2703-005509	L5006,L5007	INDUCTOR-SMD
2801-005300	OSC5000	CRYSTAL-SMD
2801-005323	OSC3000	CRYSTAL-SMD
2901-001690	C4023,C4074	FILTER-EMI SMD
2901-001786	F7000,F7001,F7002	FILTER-EMI/ESD
2901-001786	F7003,F7004,F7005	FILTER-EMI/ESD
2901-001786	F7006,F7007,F7008	FILTER-EMI/ESD
2901-001786	F7009,F7010,F7011	FILTER-EMI/ESD
2901-001786	F7012,F7013,F7014	FILTER-EMI/ESD
2904-002260	U1003	FILTER-SAW
2904-002266	U1005	FILTER-SAW
2904-002355	F3003,U3006	FILTER-SAW
2904-002357	F2001	FILTER-SAW
2904-002360	F3002	FILTER-SAW
2904-002363	F1003	FILTER-SAW
2904-002367	F1005	FILTER-SAW
2910-000386	F1001	DUPLEXER-SAW
2910-000396	F1006	DUPLEXER-SAW
2911-000308	U3000	DUPLEXER-SAW

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2911-000422	F1000	DUPLEXER-SAW
2911-000429	F2000	DUPLEXER-SAW
3003-001236	AU8000	MIC MEMS
3301-001787	L3006	BEAD-SMD
3301-001885	L6008	BEAD-SMD
3301-002078	L8009,L8010	BEAD-SMD
3301-002223	L7000,L7007	BEAD-SMD
3301-002228	L8003,L8004	BEAD-SMD
3301-002236	L6003,L6004,L8008	BEAD-SMD
3301-002238	L3012	BEAD-SMD
3301-002312	L7004,L7005,L7006	BEAD-SMD
3301-002331	L5000,L6002	BEAD-SMD
3301-002335	L8011	BEAD-SMD
3705-001937	RFS1000,RFS1001	CONNECTOR-COAXIAL
3705-001937	RFS1002,RFS1003	CONNECTOR-COAXIAL
3705-001937	RFS1004	CONNECTOR-COAXIAL
3709-001891	SIM5000	CONNECTOR-CARD EDGE
3711-007107	HEA7000	SOCKET-BOARD TO BOARD
3711-008508	HDC7001	HEADER-BOARD TO BOARD
3711-008847	BTC5000	CONNECTOR-BATTERY
3711-008995	HDC7000	HEADER-BOARD TO BOARD
3711-009058	HEA5000	HEADER-BOARD TO BOARD
3711-009073	HEA6000	HEADER-BOARD TO BOARD
3712-001604	ANT3007,ANT3008	CONNECTOR-TERMINAL
3712-001604	ANT8000,ANT8001	CONNECTOR-TERMINAL
3712-001626	ANT7000,ANT7001	CONNECTOR-TERMINAL
3712-001694	ANT1000,ANT1001	CONNECTOR-TERMINAL
3712-001694	ANT2000,ANT2001	CONNECTOR-TERMINAL
3712-001694	ANT2002,ANT2003	CONNECTOR-TERMINAL
3712-001694	ANT2004,ANT2005	CONNECTOR-TERMINAL
3712-001694	ANT3000,ANT3001	CONNECTOR-TERMINAL
3712-001694	ANT3002,ANT3003	CONNECTOR-TERMINAL
3712-001694	ANT3004,ANT3005	CONNECTOR-TERMINAL
3712-001694	ANT3006,ANT6000	CONNECTOR-TERMINAL
3712-001694	ANT6001,ANT6002	CONNECTOR-TERMINAL
3712-001694	ANT6003,ANT6004	CONNECTOR-TERMINAL
3712-001694	ANT8002,ANT8003	CONNECTOR-TERMINAL
4709-002150	F3001	COUPLER-DIRECTION
4709-002193	F1002	COUPLER-DIRECTION

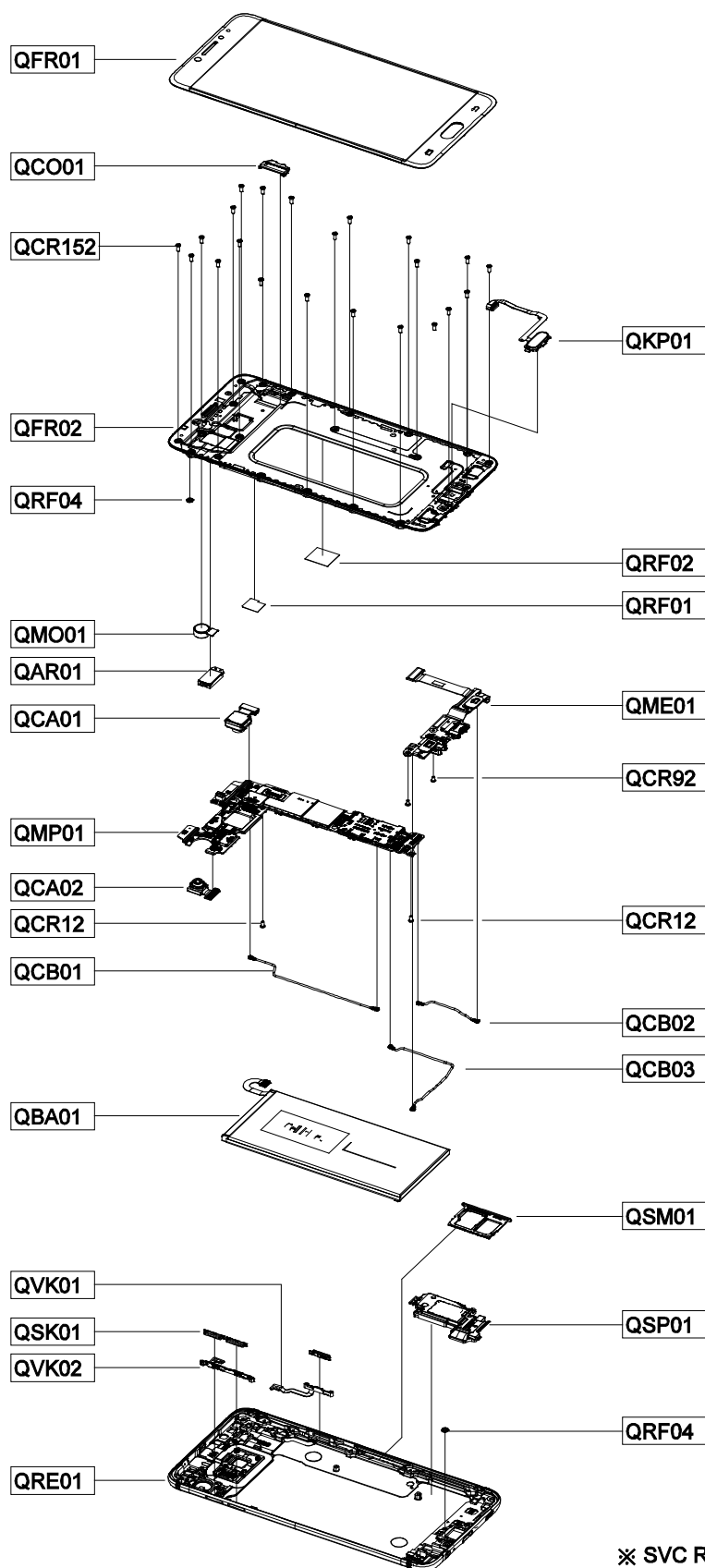
5. MAIN Electrical Parts List

4709-002278	F1004	BALUN
4709-002285	F3000	FREQ-DISTRIBUTER
4709-002351	U3001	FREQ-DISTRIBUTER
GH60-00009A	CON1002,CON1003	CLIP-SM-G935A
GH60-00009A	CON1004	CLIP-SM-G935A
GH61-07836A	CON1001	CLIP
GH61-09798A	SUS2000	PLATE-PCB
GH62-00040A	CON7000,CON7001	PAD GAP-SMD GASKET (SM-G925F)
GH63-12874A	SC3000	SHIELD-BOT_APT
GH63-13767A	SC8007	SHIELD-TOP_RCV
GH63-13768A	SC8006	SHIELD-BOT_AUDIO
GH63-13769A	SC8005	SHIELD-BOT_PMIC
GH63-13770A	SC8001	SHIELD-BOT_RF
GH63-13771A	SC8004	SHIELD-TOP_AP
GH02-14178A	SC8004_UP1	THERMAL CONDUCTIVE SHEET-AP CU SHEET
GH02-14179A	SC8004_UP2	THERMAL CONDUCTIVE SHEET-ME MORY CU SHEET
GH63-13772A	SC8000	SHIELD-TOP_MAIN CHARGER IC
GH63-13773A	SC8003	SHIELD-TOP_SUB CHARGER IC
GH63-13774A	SC8002	SHIELD-TOP_WIFI

Please consult the GSPN website (Samsung Portal) for the most recent version of the product's part list.

4. Exploded View and Parts List

4-1. Cellular phone Exploded View



※ SVC REPAIR TAPE
QRT01, QRT02, QRT03
QRT04, QRT05, QRT06

2. Specification

2-1. CDMA&GPS General Specification

ITEM	CDMA	GPS
Tx Freq. range	824.025~848.985MHz	-
Rx Freq. range	869.025~893.985MHz	1575.42MHz
Channel Bandwidth	1.23MHz	2MHz
Channel Spacing	30KHz	Not Used
Number of Channel	832	1
Duplex Separation	45MHz	-
In/Output Impedance	50Ohm	50Ohm
Tx Local Frequency	$F_{TX} * 1.6666$	-
Rx Local Frequency	$F_{RX} * 2$	$F_{RX} * 2$
TCXO Frequency	19.2MHz	19.2MHz
Freq. Stability	(FRX-45MHz)±300Hz	-
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃

2. Specification

Item		GSM 850	EGSM 900	DCS1800	PCS1900
Freq. Band[MHz] Uplink/Downlink		824~849 869~894	880~915 925~960	1710~1785 1805~1880	1850~1910 1930~1990
ARFCN range		128~251	0~124 & 975~1023	512~885	512~810
Tx/Rx spacing		45MHz	45MHz	95MHz	80MHz
Mod. Bit rate/ Bit Period		270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us
Time Slot Period/ Frame Period		576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms
Modulation	GSM/ GPRS	0.3GMSK	0.3GMSK	0.3GMSK	0.3GMSK
MS Power		33dBm ~5dBm	33dBm ~5dBm	30dBm ~0dBm	30dBm ~0dBm
Power Class		5pcl ~ 19pcl	5pcl ~ 19pcl	0pcl ~ 15pcl	0pcl ~ 15pcl
Sensitivity		-102dBm	-102dBm	-100dBm	-100dBm
TDMA Mux		8	8	8	8
Cell Radius		35Km	35Km	2Km	2Km

2. Specification

2-2. GSM Tx Power Class

TX Power control level	GSM850	TX Power control level	EGSM900	TX Power control level	DCS1800	TX Power control level	PCS1900
5	33±2 dBm	5	33±2 dBm	0	30±3 dBm	0	30±3 dBm
6	31±2 dBm	6	31±2 dBm	1	28±3 dBm	1	28±3 dBm
7	29±2 dBm	7	29±2 dBm	2	26±3 dBm	2	26±3 dBm
8	27±2 dBm	8	27±2 dBm	3	24±3 dBm	3	24±3 dBm
9	25±2 dBm	9	25±2 dBm	4	22±3 dBm	4	22±3 dBm
10	23±2 dBm	10	23±2 dBm	5	20±3 dBm	5	20±3 dBm
11	21±2 dBm	11	21±2 dBm	6	18±3 dBm	6	18±3 dBm
12	19±2 dBm	12	19±2 dBm	7	16±3 dBm	7	16±3 dBm
13	17±2 dBm	13	17±2 dBm	8	14±3 dBm	8	14±3 dBm
14	15±2 dBm	14	15±2 dBm	9	12±4 dBm	9	12±4 dBm
15	13±2 dBm	15	13±2 dBm	10	10±4 dBm	10	10±4 dBm
16	11±3 dBm	16	11±3 dBm	11	8±4 dBm	11	8±4 dBm
17	9±3 dBm	17	9±3 dBm	12	6±4 dBm	12	6±4 dBm
18	7±3 dBm	18	7±3 dBm	13	4±4 dBm	13	4±4 dBm
19	5±3 dBm	19	5±3 dBm	14	2±5 dBm	14	2±5 dBm
				15	0±5 dBm	15	0±5 dBm

2. Specification

2-3. WCDMA General Specification

	WCDMA2100	WCDMA1900	WCDMA850	WCDMA900
Freq. Band[MHz] Uplink/Downlink	1922~1977 2112~2167	1852~1907 1932~1987	824~849 869~894	880~915 925~960
ARFCN range	UL: 9612~9888 DL: 10562~10838	UL: 9262~9538 DL: 9662~9938	UL: 4132~4233 DL: 4357~4458	UL: 2712~2863 DL: 2937~3088
Tx/Rx spacing	190MHz	80MHz	45MHz	45MHz
Mod. Bit rate/ Bit Period	3.84 Mcps	3.84 Mcps	3.84 Mcps	3.84 Mcps
Time Slot Period/Frame Period	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms
Modulation	QPSK/HQPSK	QPSK/HQPSK	QPSK/HQPSK	QPSK/HQPSK
MS Power	24dBm~-50dBm	24dBm~-50dBm	24dBm~-50dBm	24dBm~-50dBm
Power Class	3(max+24dBm)	3(max+24dBm)	3(max+24dBm)	3(max+24dBm)
Sensitivity	-106.7dBm	-106.7dBm	-106.7dBm	-106.7dBm
TDMA Mux	8	8	8	8
Cell Radius	2Km	2Km	2Km	2Km

2. Specification

2-4. TD-SCDMA General Specification

	Band34	Band39
Freq. Band[MHz] Uplink/Downlink	2010~2025	1880~1920
ARFCN range	10054~10121	9404~9596
Tx/Rx spacing	/	/
TCXO Frequency	19.2MHz	19.2MHz
Modulation	QPSK/16-QAM	QPSK/16-QAM
MS Power	-49dBm~ 25dBm	-49dBm~ 25dBm
Power Class	2 (max: 24 +1/-3dBm)	2 (max: 24 +1/-3dBm)
Sensitivity	-108dBm/1.28MHz	-108dBm/1.28MHz
Cell Radius	>10Km	>10Km
In/Output Impedance	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃

2. Specification

2-5. LTE General Specification (FDD)

	Band1 (FDD)	Band3 (FDD)	Band5 (FDD)	Band7 (FDD)	Band8 (FDD)
Freq. Band[MHz] UL/DL	1920~1980 2110~2170	1710~1785 1805~1880	824~849 869~894	2500~2570 2620~2690	880~915 925~960
ARFCN range UL/DL	18000~18599 0~599	19200~19949 1200~1949	20400~20649 2400~2649	20750~20449 2750~3449	21450~21799 3450~3799
Tx/Rx spacing	190MHz	95MHz	45MHz	120MHz	45MHz
TCXO Frequency	19.2MHz	19.2MHz	19.2MHz	19.2MHz	19.2MHz
Modulation	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM
MS Power	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm
Power Class	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)
Sensitivity	-99.3(BW : 5 MHz) -96.3(BW : 10 MHz) -94.5(BW : 15MHz) -93.3(BW : 20MHz)	-101(BW : 1.4MHz) -98(BW : 3MHz) -96.3(BW : 5 MHz) -93.3(BW : 10 MHz) -91.5(BW : 15MHz) -90.3(BW : 20MHz)	-102.5(BW:1.4MHz) -99.5(BW : 3MHz) -97.3(BW : 5 MHz) -94.3(BW : 10 MHz)	-97.3(BW : 5 MHz) -94.3(BW : 10 MHz) -92.5(BW : 15MHz) -91.3(BW : 20MHz)	- 101.5(BW : 1.4MHz) -98.5(BW : 3MHz) -96.3 (BW : 5 MHz) -93.3(BW : 10 MHz)
Cell Radius	>5Km	>5Km	>5Km	>5Km	>5Km
In/Output Impedance	50Ω	50Ω	50Ω	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃

2. Specification

	Band38 (TDD)	Band39 (TDD)	Band40 (TDD)	Band41 (TDD)
Freq. Band[MHz] Uplink/Downlink	2570~2620	1880~1920	2300~2400	2555~2655
ARFCN range	37750~38249	38250~38649	38650~39649	40240~41240
Tx/Rx spacing	/	/	/	/
TCXO Frequency	19.2MHz	19.2MHz	19.2MHz	19.2MHz
Modulation	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM
MS Power	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm
Power Class	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)
Sensitivity	-99.3(BW : 5 MHz) -96.3(BW : 10 MHz) -94.5(BW : 15MHz) -93.3(BW : 20MHz)	-99.3(BW : 5 MHz) -96.3(BW : 10 MHz) -94.5(BW : 15MHz) -93.3(BW : 20MHz)	-99.3(BW : 5 MHz) -96.3(BW : 10 MHz) -94.5(BW : 15MHz) -93.3(BW : 20MHz)	-97.3(BW : 5 MHz) -94.3(BW : 10 MHz) -92.5(BW : 15MHz) -91.3(BW : 20MHz)
Cell Radius	>5Km	>5Km	>5Km	>5Km
In/Output Impedance	50Ω	50Ω	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃

3. Operation Instruction and Installation

Main Function

Item	Description
OS	Android V6.0
RF	GSM850 /GSM900/ DCS1800/ PCS1900 CDMA:BC0 WCDMA: B1/ B2/ B5/ B8 TDSCDMA: B34/B39 LTE: B1/ B3/ B5/B7/ B8/ B38/ B39/ B40/ B41
Battery	3300mAh
Base Band	Octa-Core 2.2Ghz
Other RF	GPS, Glonass, Beidou, BT4.2, USB Type-C, WIFI 802.11 a/b/g/n, NFC,FM
Camera	16MP Main CAM; 16MP Sub CAM
LCD	5.7." super AMOLED, 1920*1080(FHD)
RAM	4GB
Sensor	Accelerometer, Proximity, RGB light sensor, Fingerprint sensor, Geomagnetic sensor
Accessory	Charger: 9V/1.67A Data cable: 2.8pi, 1.2m

9. Reference Abbreviate

Reference Abbreviate

- **AAC**: Advanced Audio Coding.
- **AVC** : Advanced Video Coding.
- **BER** : Bit Error Rate
- **BPSK**: Binary Phase Shift Keying
- **CA** : Conditional Access
- **CDM** : Code Division Multiplexing
- **C/I** : Carrier to Interference
- **DMB** : Digital Multimedia Broadcasting
- **EN** : European Standard
- **ES** : Elementary Stream
- **ETSI**: European Telecommunications Standards Institute
- **MPEG**: Moving Picture Experts Group
- **PN** : Pseudo-random Noise
- **PS** : Pilot Symbol
- **QPSK**: Quadrature Phase Shift Keying
- **RS** : Reed-Solomon
- **SI** : Service Information
- **TDM** : Time Division Multiplexing
- **TS** : Transport Stream

1. Safety Precautions

1-1. Repair Precaution

Before attempting any repair or detailed tuning, shield the device from RF noise or static electricity discharges.

Use only demagnetized tools that are specifically designed for small electronic repairs, as most electronic parts are sensitive to electromagnetic forces.

Use only high quality screwdrivers when servicing products. Low quality screwdrivers can easily damage the heads of screws.

Use only conductor wire of the properly gauge and insulation for low resistance, because of the low margin of error of most testing equipment.

We recommend 22-gauge twisted copper wire.

Hand-soldering is not recommended, because printed circuit boards (PCBs) can be easily damaged, even with relatively low heat. Never use a soldering iron with a power rating of more than 100 watts and use only lead-free solder with a melting point below 250°C (482°F).

Prior to disassembling the battery charger for repair, ensure that the AC power is disconnected. Always use the replacement parts that are registered in the SEC system. Third-party replacement parts may not function properly.

1. Safety Precautions

1-2. ESD(Electrostatically Sensitive Devices) Precaution

Many semiconductors and ESDs in electronic devices are particularly sensitive to static discharge and can be easily damaged by it. We recommend protecting these components with conductive anti-static bags when you store or transport them.

Always use an anti-static strap or wristband and remove electrostatic buildup or dissipate static electricity from your body before repairing ESDs.

Ensure that soldering irons have AC adapter with ground wires and that the ground wires are properly connected.

Use only desoldering tools with plastic tips to prevent static discharge.

Properly shield the work environment from accidental electrostatic discharge before opening packages containing ESDs.

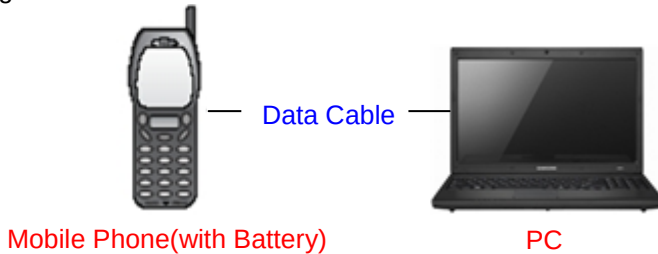
The potential for static electricity discharge may be increased in low humidity environments, such as air-conditioned rooms. Increase the airflow to the working area to decrease the chance of accidental static electricity discharges.

6. Level 1 Repair

6-1. S/W Download

6-1-1. Prepare for S/W Downloading

- Diagram of connection



6-1-2. How to download S/W

1) Downloading Binary Files

- Binary file for downloading SM-C7010
 - [xxx.pit](#)
 - [BL_XXXX.tar.md5](#)
 - [AP_XXXX.tar.md5](#)
 - [CP_XXXX.tar.md5](#)
 - [CSC_xxxx.tar.md5](#)

(file size is about 3.8GB)

2) Prepare for Downloading

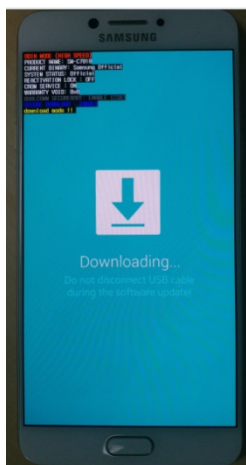
- Downloader Program ([Odin3 v3.10.7exe](#))
- SM-C7010 Mobile Phone
- Data Cable
- Binary file

3) Boot the mobile phone by pressing 'Home + Vol Down + Power key at the same time, If you do properly, you can see the following message on the main LCD as the following.



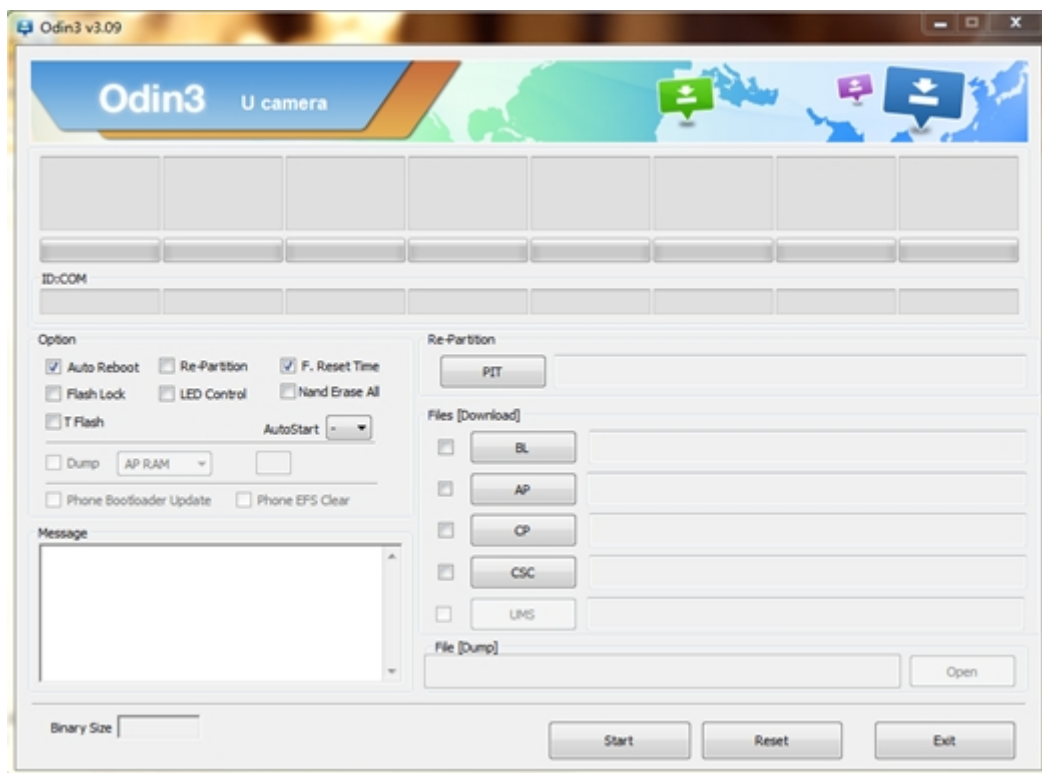
6. Level 1 Repair

4) Press the Vol Up Key again, and you will see below message on Main LCD.



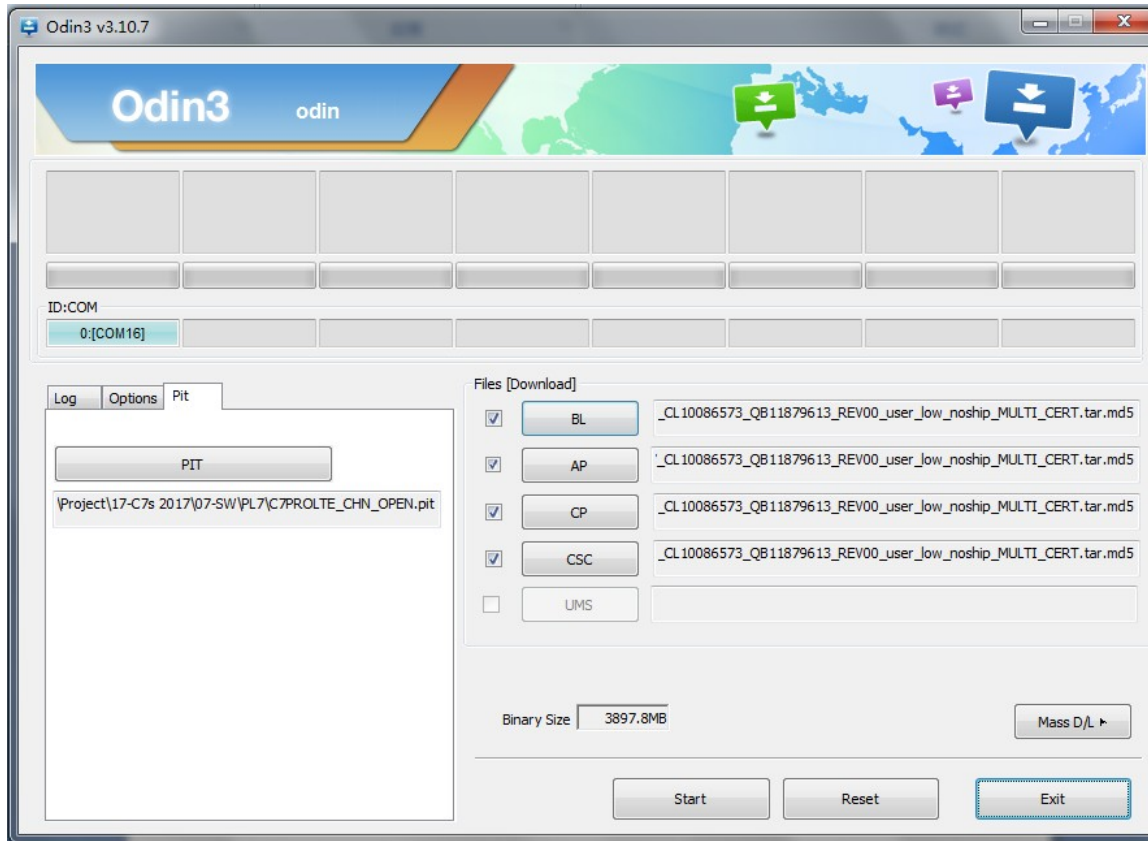
“ Downloading...” ”

5) Load the binary download program.



6. Level 1 Repair

6) Connect mobile and computer. The program show as follow.



Select the file as above:

- BL_XXXX.tar.md5
- AP_XXXX.tar.md5
- CP_XXXX.tar.md5
- CSC_xxxx.tar.md5

7) Now press the button "Start".

8) Now it's time to take a rest and finish the downloading.

9) After finished downloading of phone binary, the mobile phone will restart automatically.

※ **Caution. Never disconnect during the S/W downloading.**

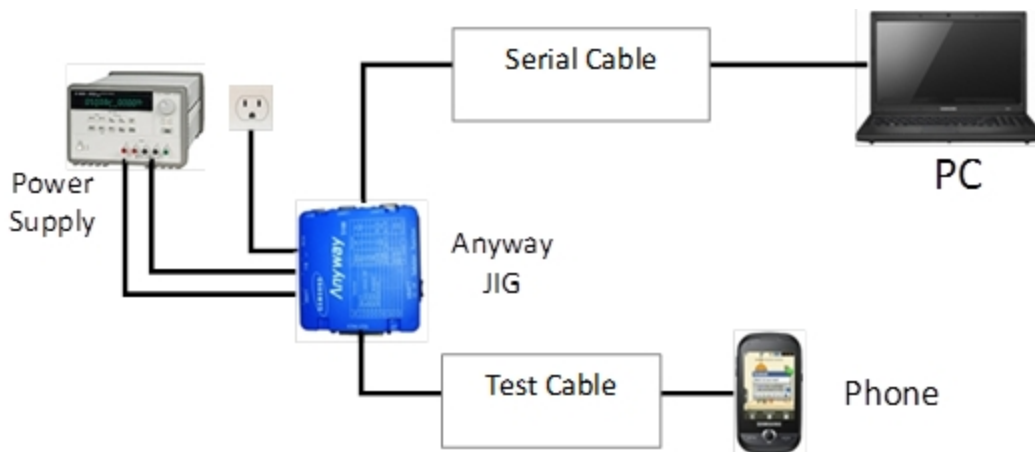
6. Level 1 Repair

6-2 IMEI writing

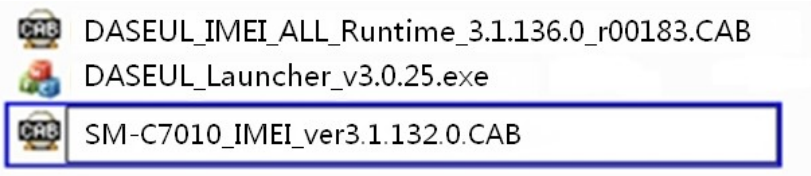
6-2-1 Preparation

- New IMEI writing Program has been released.
- Supported Model: Models which CAB files are uploaded on HHP svc INI File category, instead of ini file.
- Refer to below IMEI writing procedure.

-H/W



-S/W

① Library Install	To use Daseul, library files should be installed. Refer to SVC Bulletin “(11-82) Daseul(New IMEI writing Program) Library Install guide_rev1.0”
② Launcher	DASEUL_SVC_Launcher_v3_0_25 or higher -Uploaded on HHPsvc Notice
③ Runtime File	1. DASEUL_IMEI_ALL_Runtime_3.1.136_r00183 .CAB or higher -Uploaded on HHPsvc Notice 2. Make 'ModelName' folder at the same position with launcher & Runtime file. 
④ Model File	Copy Model File under the 'ModelName' folder

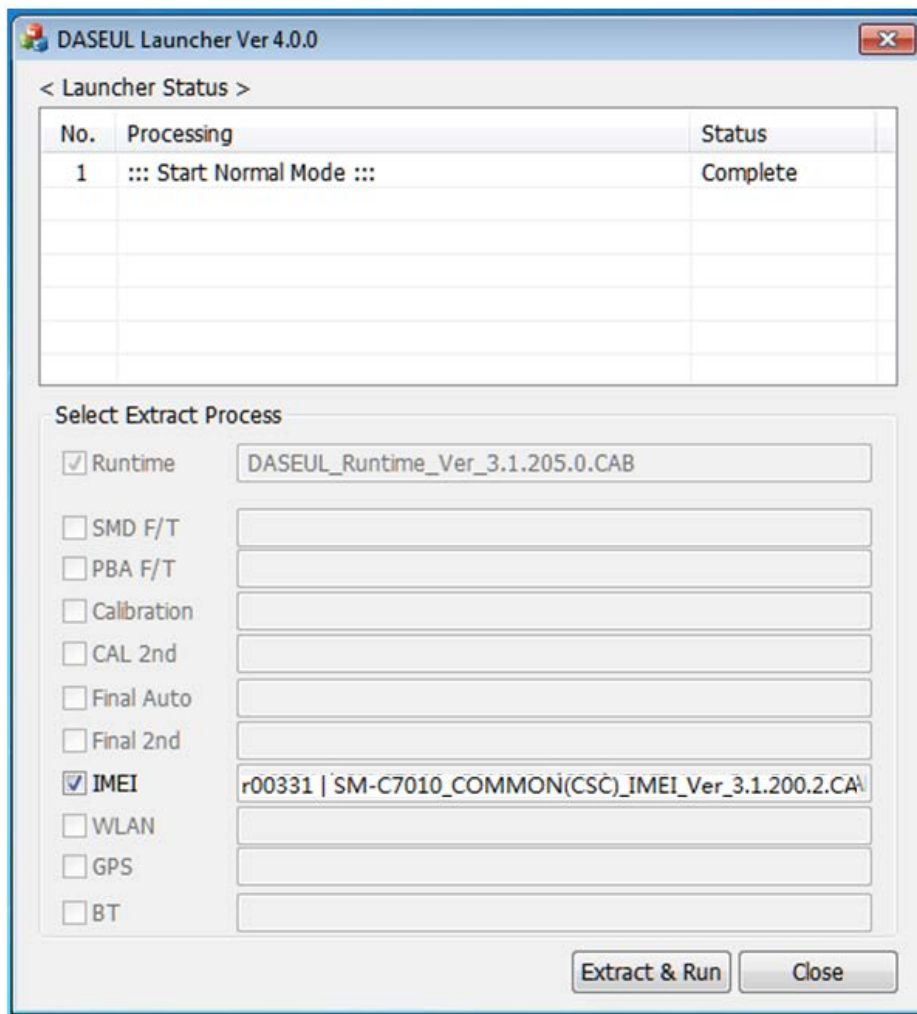
6. Level 1 Repair

6-2-2 IMEI writingProcess

1. Run DASEUL_Launcher_v4.0.0

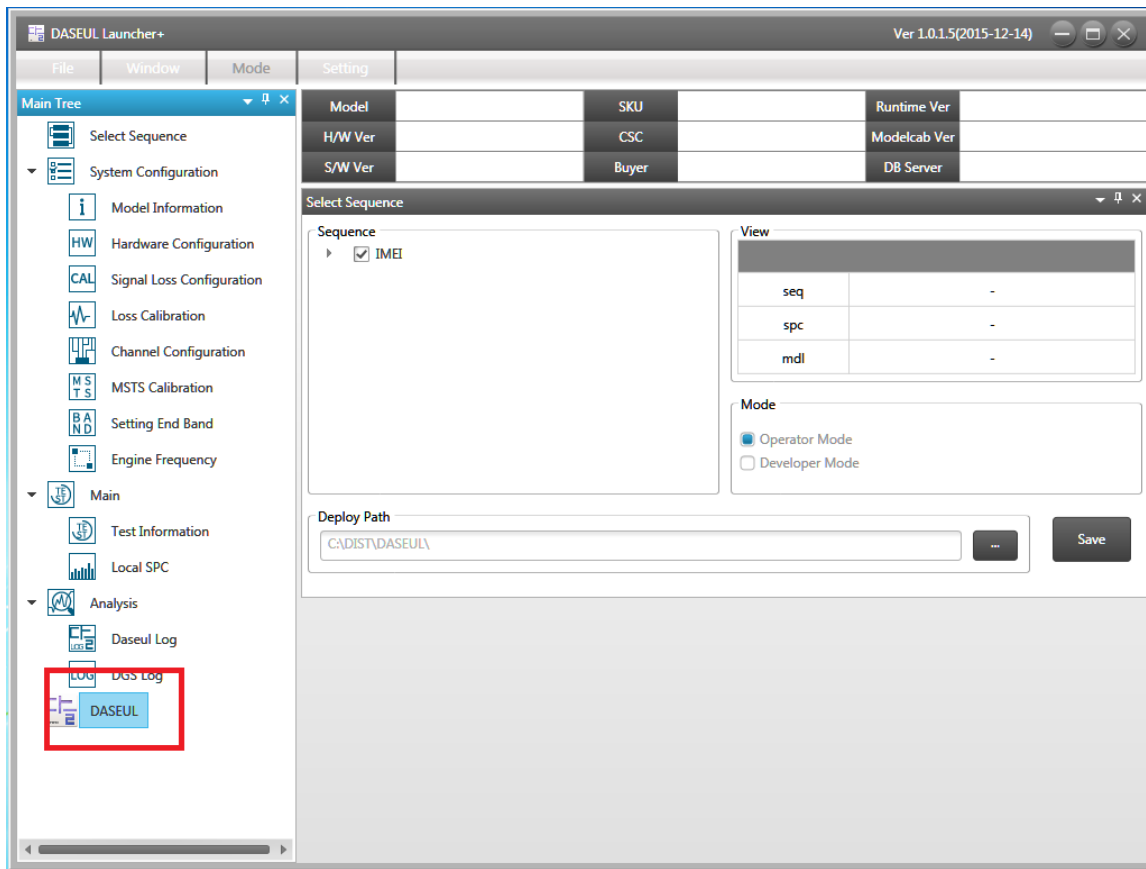
 DASEUL_IMEI_ALL_SVC_Runtime_3.1.173.0_r00270.CAB
 DASEUL_Runtime_SVC_Ver_3.1.173.0.CAB
 DASEUL_SVC_Launcher_v3.0.12.exe

2. Select Service Mode

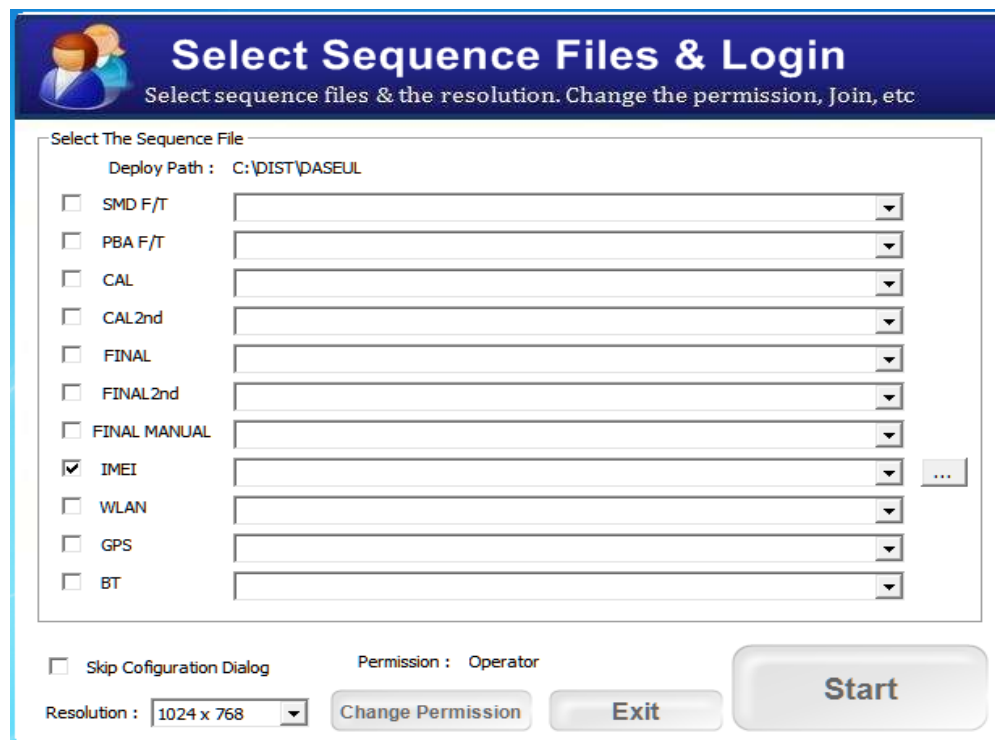


6. Level 1 Repair

3. Click DASEUL Select folder where the Launcher exists

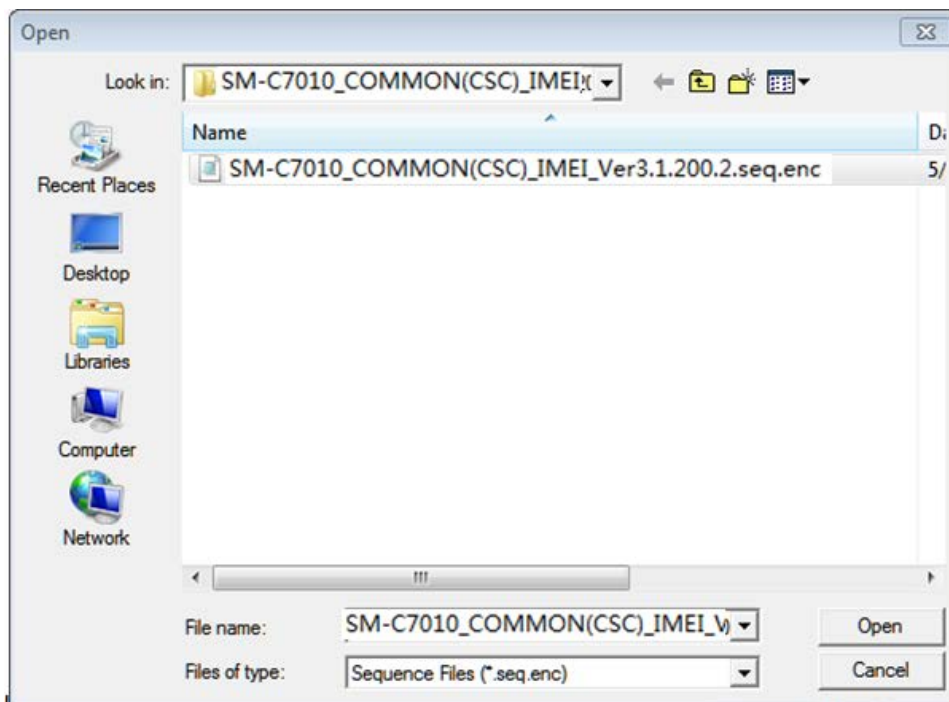


4. Select folder where the Launcher exists

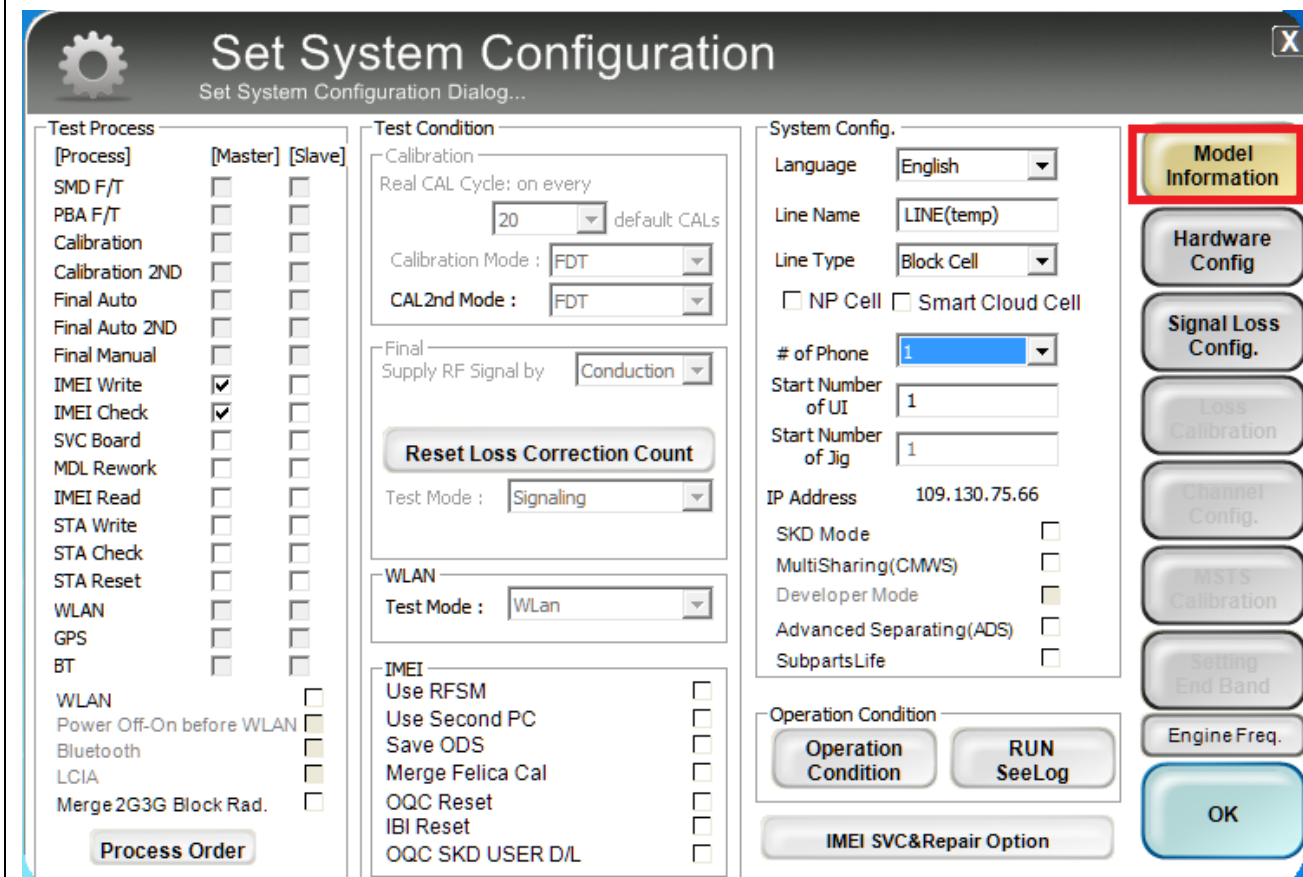


6. Level 1 Repair

5. Select Mode

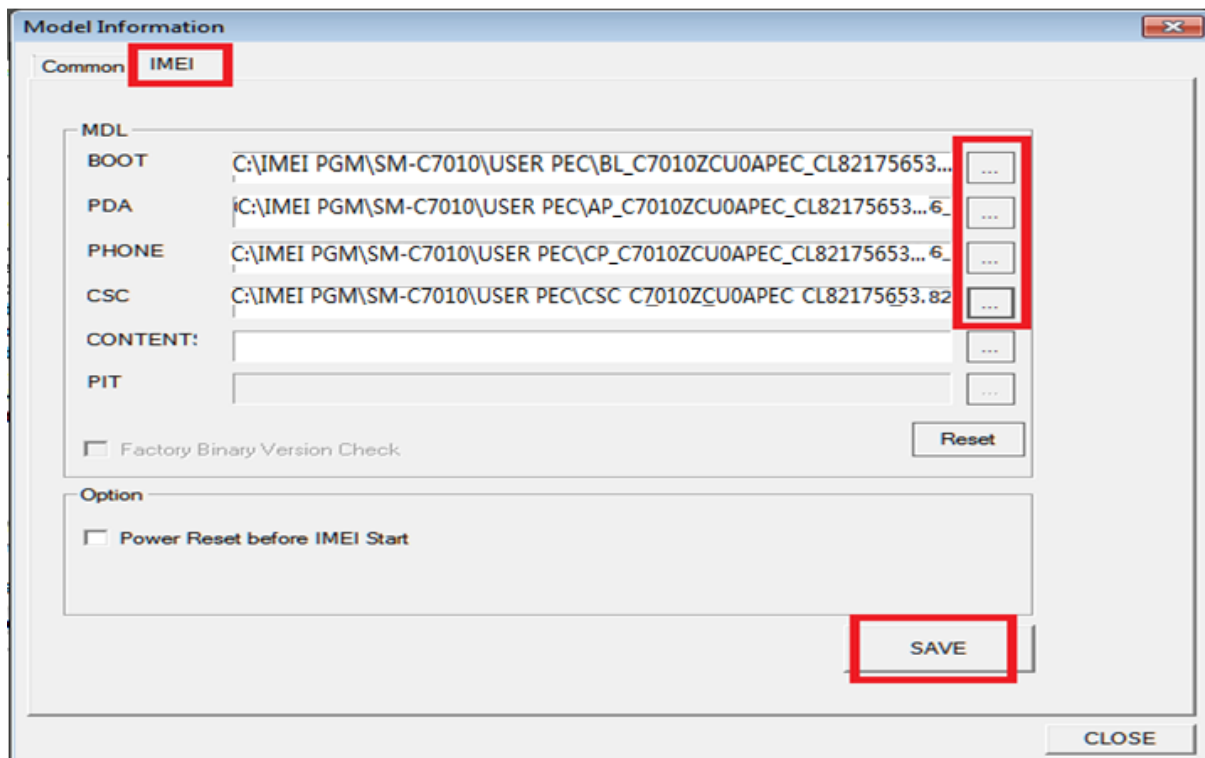


6. Check 'Model information'



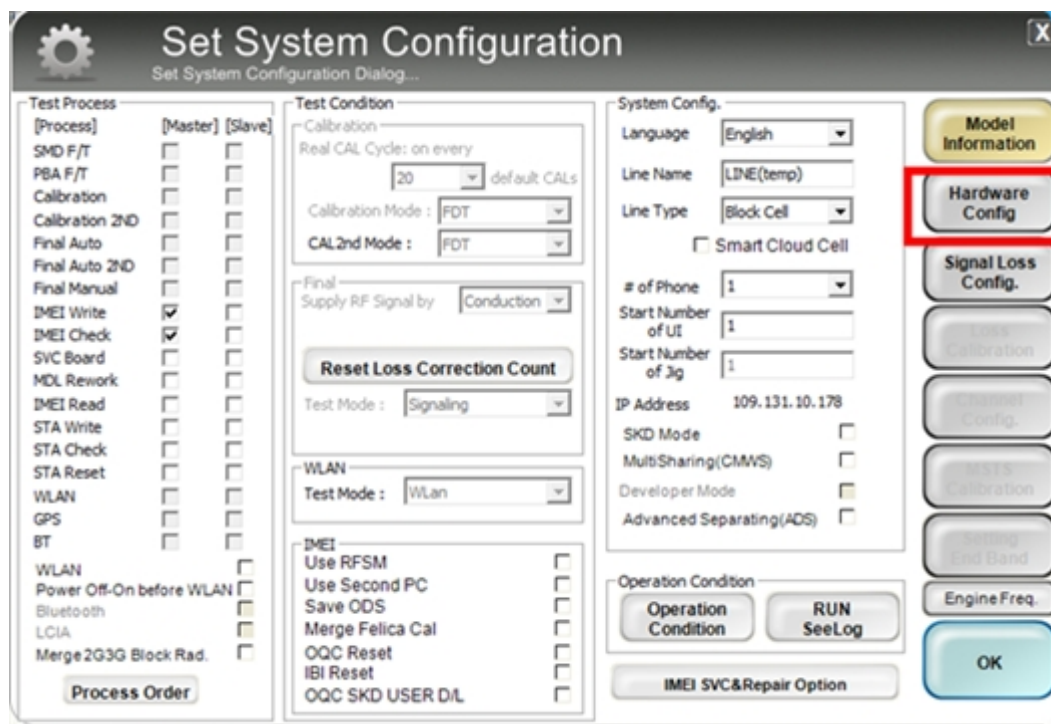
6. Level 1 Repair

7. choose download binary



8. Click IMEI, Then choose binary, click SAVE.

9. Click 'Hardware Config'



6. Level 1 Repair

10. Click 'Port Setting'

The screenshot shows the 'Hardware Component Configuration' window. The 'Phone' section on the left has a 'Port Setting' button highlighted with a red box. Other sections include 'MSTS Sharing Controller', 'DEMS', 'Barcode Reader', 'PBA F/T', 'Power Detector', 'HDMI Type', 'HDMI JIG', 'SMD F/T', 'I/F Address', 'AP(2.4G) SSID', 'AP(5G) SSID', 'GPS Generator', 'WLAN', and 'Label Printer'. Each section has various configuration options and 'Port Setting' buttons.

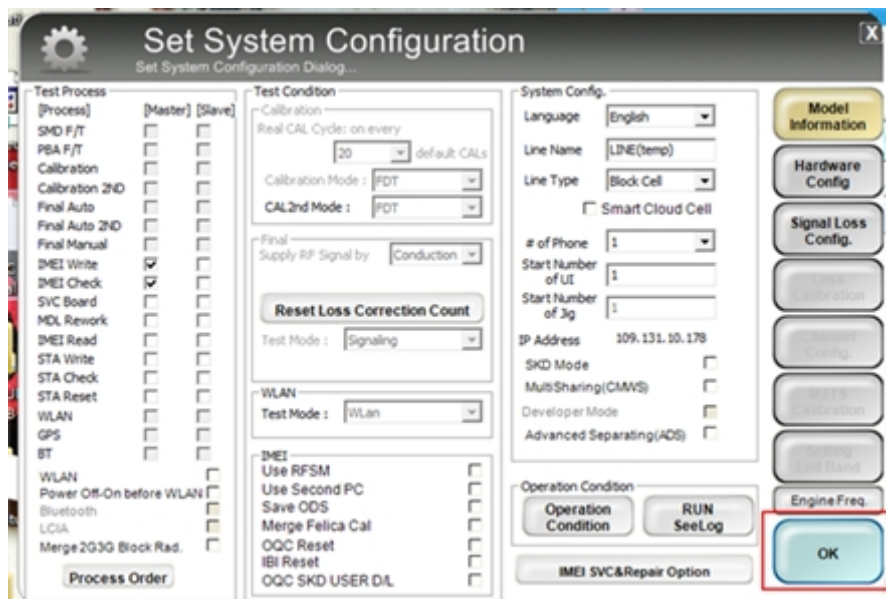
11. Select Port Number and SAVE

The screenshot shows the 'Hardware Component Configuration' window with the 'Phone IO Bus Setting' dialog box open. The dialog box has a 'Common' section with 'BaudRate' (115200), 'Data Bit' (8), 'Parity' (No), and 'Stop Bit' (1). It also has a table for 'Port #1' and 'Port #2' settings. The 'Port #1' dropdown is highlighted with a red box and labeled 'serial port'. The 'Port #2' dropdown is highlighted with a red box and labeled 'binary DL port'. The 'SAVE' button is also visible.

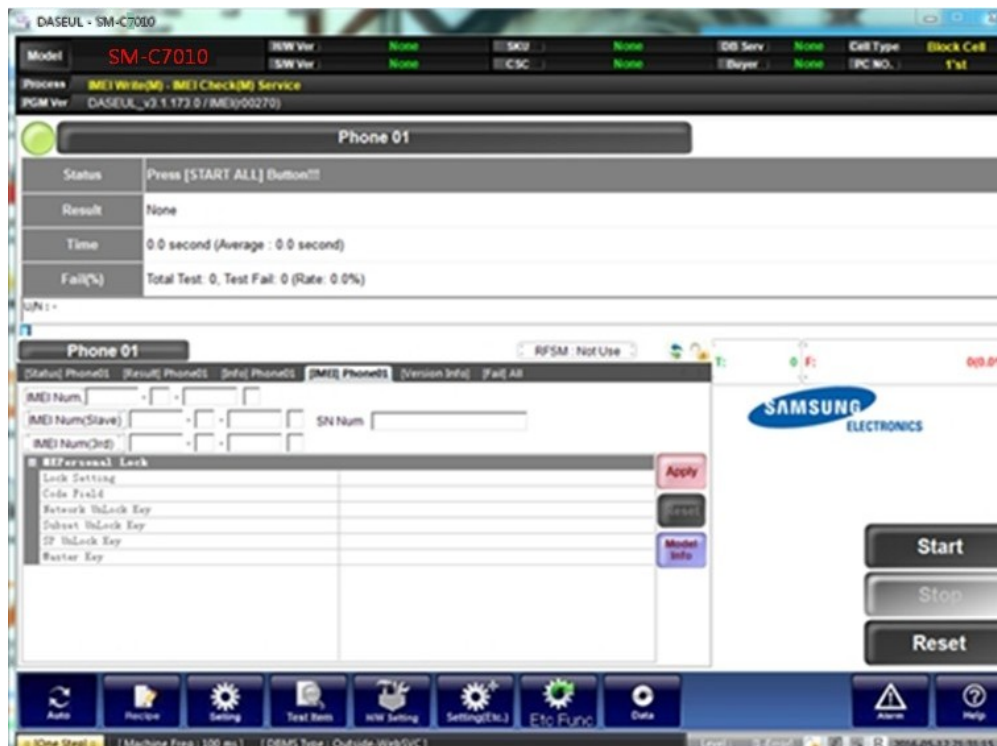
No.	Port #1	Port #2
1	4	13

6. Level 1 Repair

12. Click OK to proceed



13. Click Model Info and OK when pop-up shows

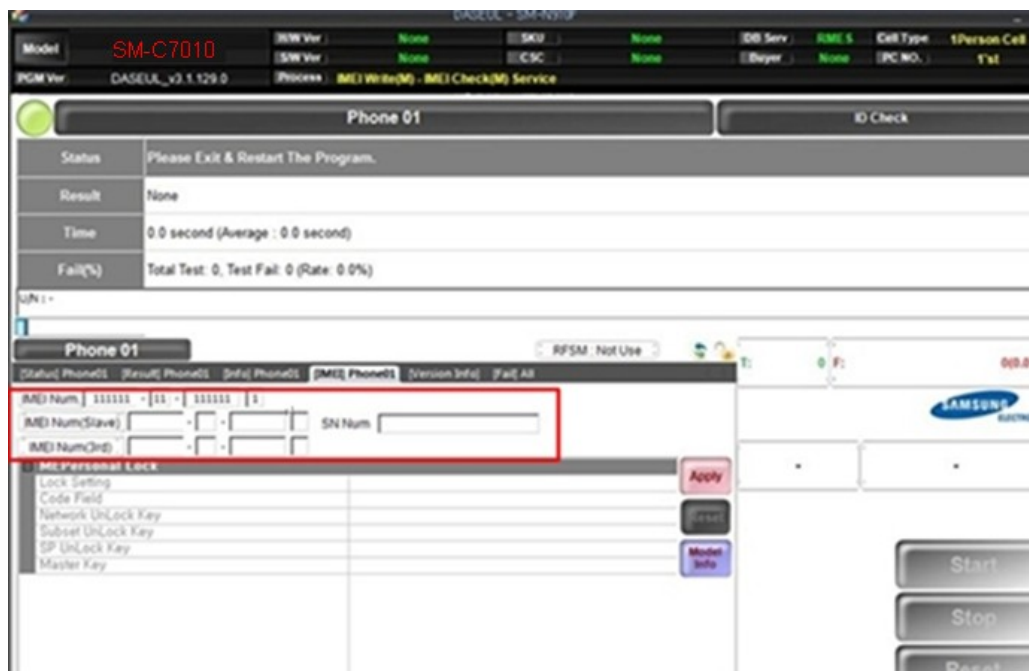


6. Level 1 Repair

14. ClickOK

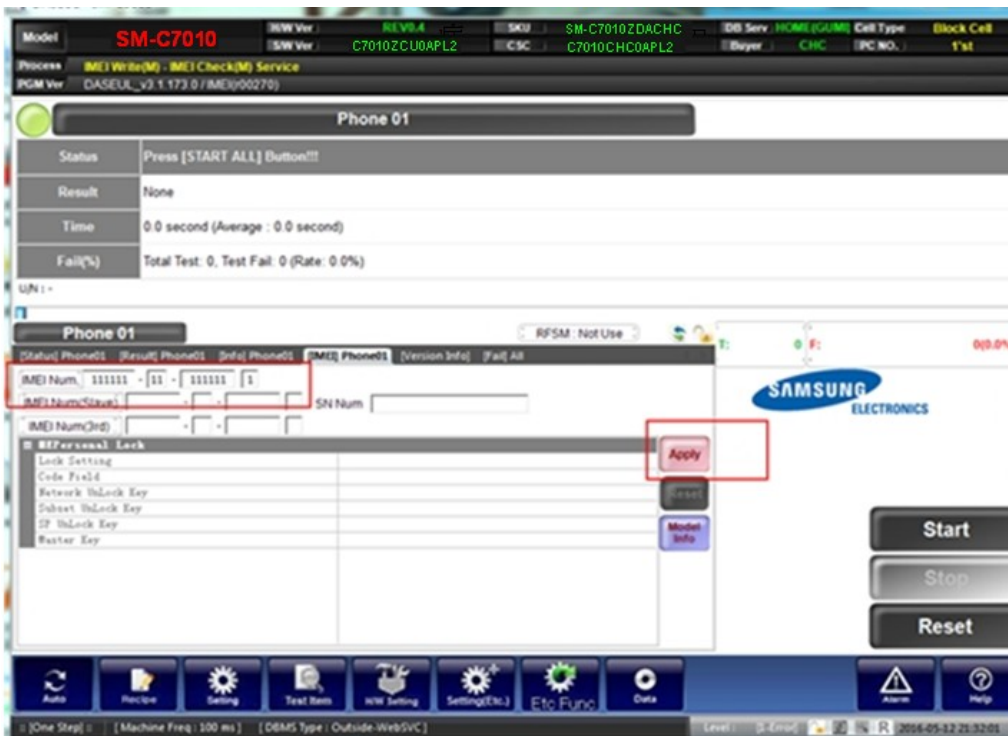


15. Input IMEI Number and click apply.



6. Level 1 Repair

16 ClickStart ALL

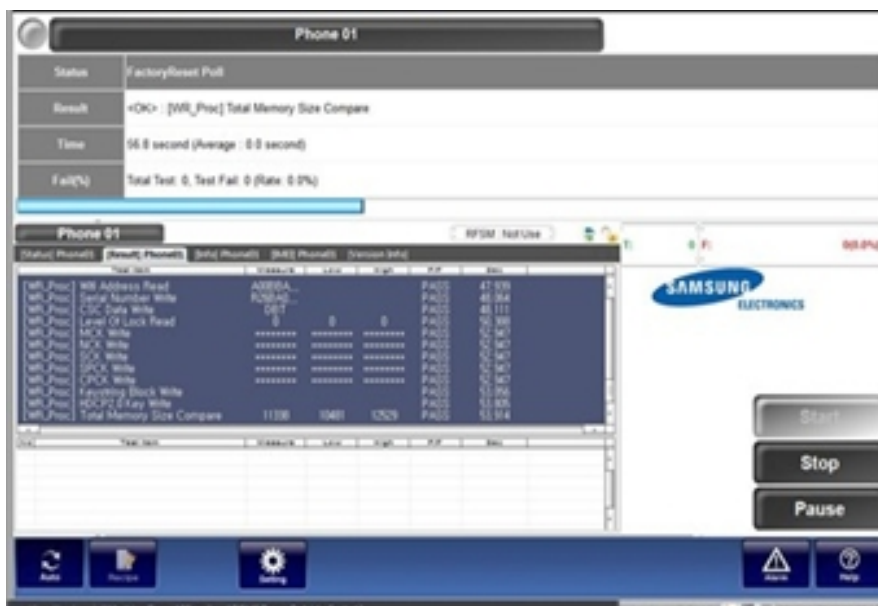


17. Connect the phone to Anyway JIG

※ When you connect the phone, the phone should be turned off.

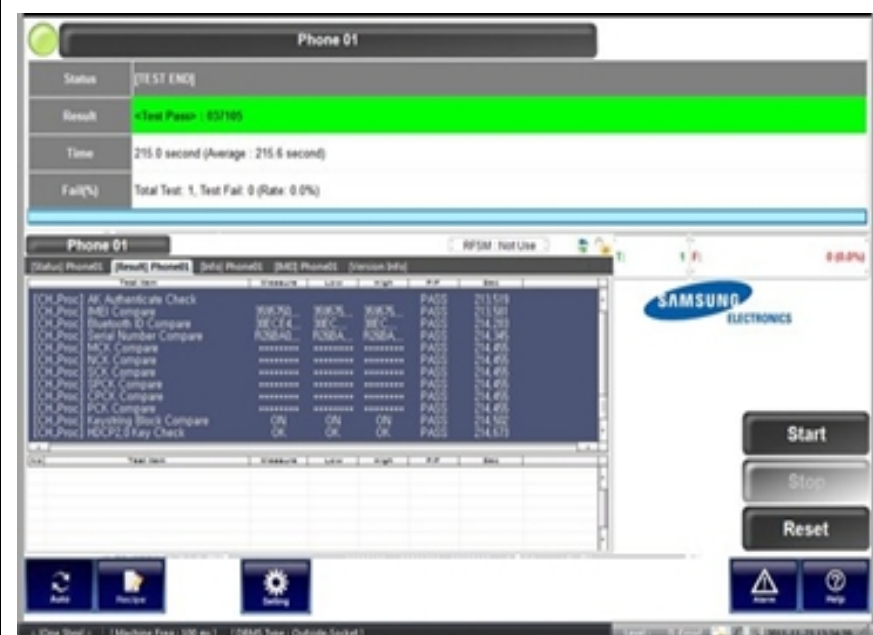
After connecting the phone, the phone will be booted automatically.

18. IMEI Writing Proceeding



6. Level 1 Repair

19. IMEI Writing Success



6. Level 1 Repair

6-3.RF Calibration

6-3-1. Required items in order to calibrate RF

- Installation program :RF Calibration Program
 - Daseul_Launcher_vx.x.xx.exe
 - Daseul_CAL_ALL_Runtime_x.x.xxx.x.CAB
 - Model File (**Slave: SM-C7010_OPEN_CALIBRATION_Ver_3.1.260.2_S; Master: SM-C7010_OPEN_CALIBRATION_Ver_3.1.260.2_M**)

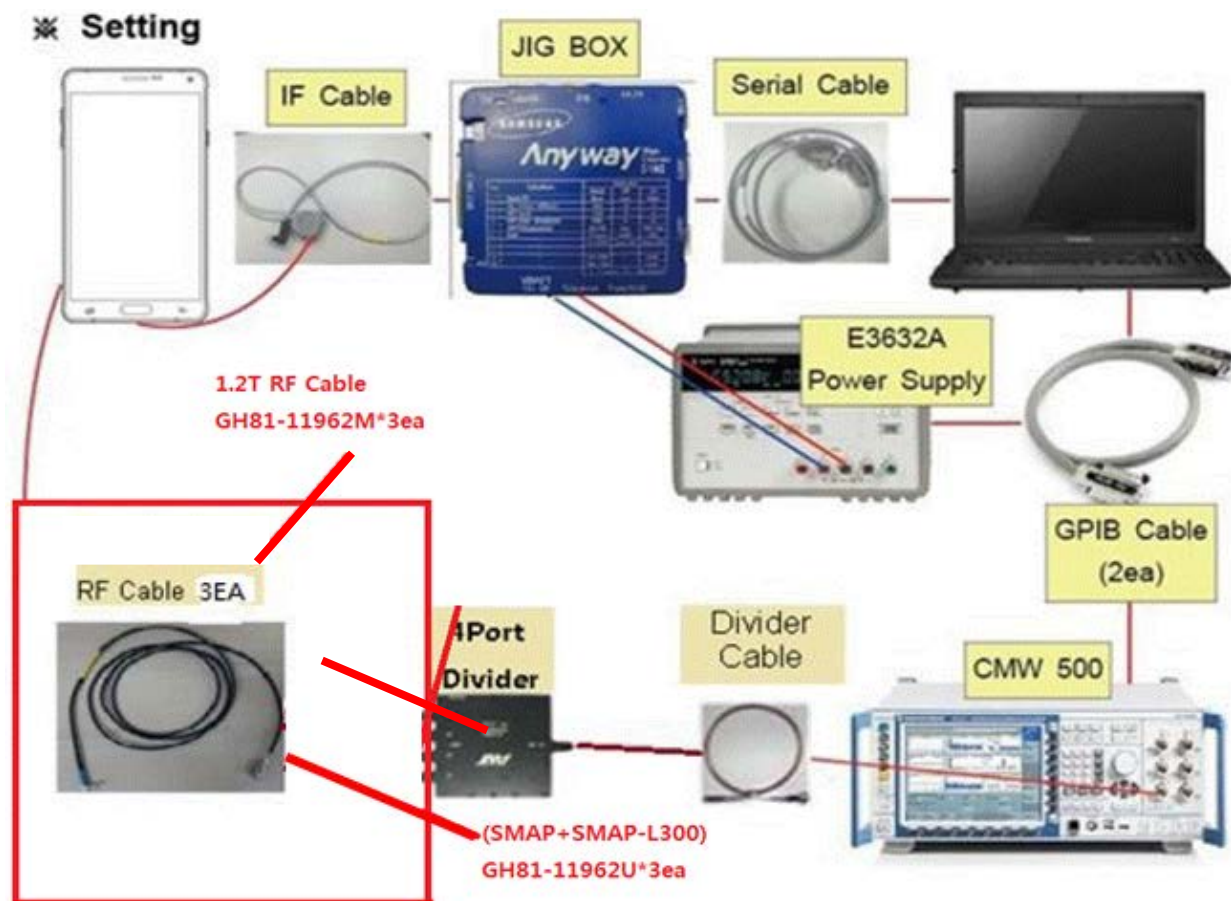
※ It is required to use the latest program.

- **SM-C7010** Mobile Phone
- E3632A Power Supply
- JIG BOX (Anyway S103)
- Adapter (GH81-11888K)
- RF Cable(GH81-11962U) 3EA
- 4Port Divider(GH81-11962A 1EA)
- R&S CMW500
- GPIB Cable(2ea)
- IF Cable(GH81-11962W)
- UART Serial Cable
- 1.35ΦRF Cable (GH81-11962M 3ea)
- Divider Cable(GH81-11962B 1EA)

- Table of test cables

IF Cable	GH81-11962W			
	USB C type			
RF Cable (Manual)	GH81-11962M	GH81-11962U		
	1.2T, 102mm 3ea 	SMAP, 300mm 3ea 		
4 Port Divider	GH81-11962A	GH81-11962B	GH81-11962E	
	Divider 1ea 	Divider Cable 1ea 	50Ω terminator 1ea 	

6. Level 1 Repair



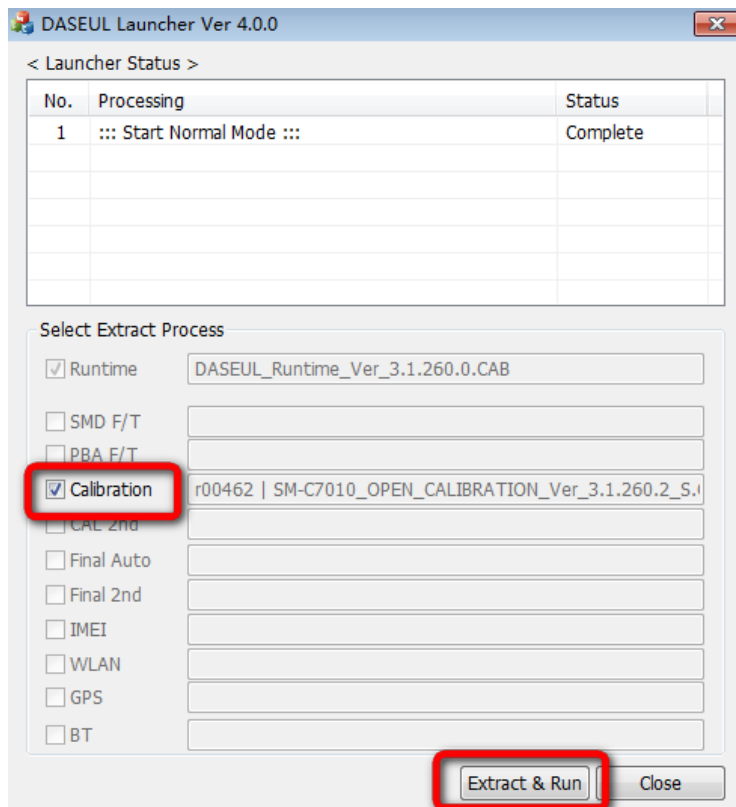
6. Level 1 Repair

6-3-2. RF Calibration Program

1. Run the RF Calibration Program Launcher, '[DASEUL_Launcher_vx.x.xx.exe](#)'.

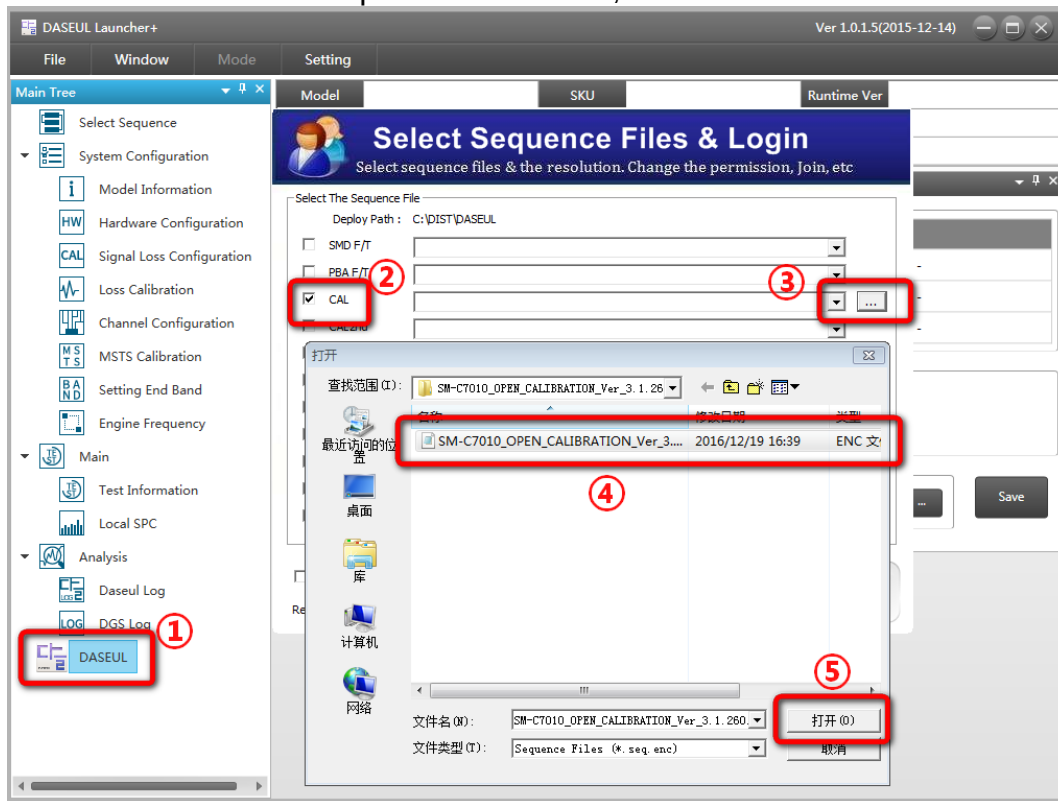
	DASEUL_CAL_ALL_Runtime_3.1.260.0_...	2016/12/20 14:33	ALZip cab 文件	128,382 KB
	DASEUL_Launcher_v4.0.0	2016/12/20 14:32	应用程序	5,896 KB
	SM-C7010_OPEN_CALIBRATION_Ver_...	2016/12/20 13:14	ALZip cab 文件	463 KB

2. Check the '[Calibration](#)' menu, and select '[Extract & Run](#)'.

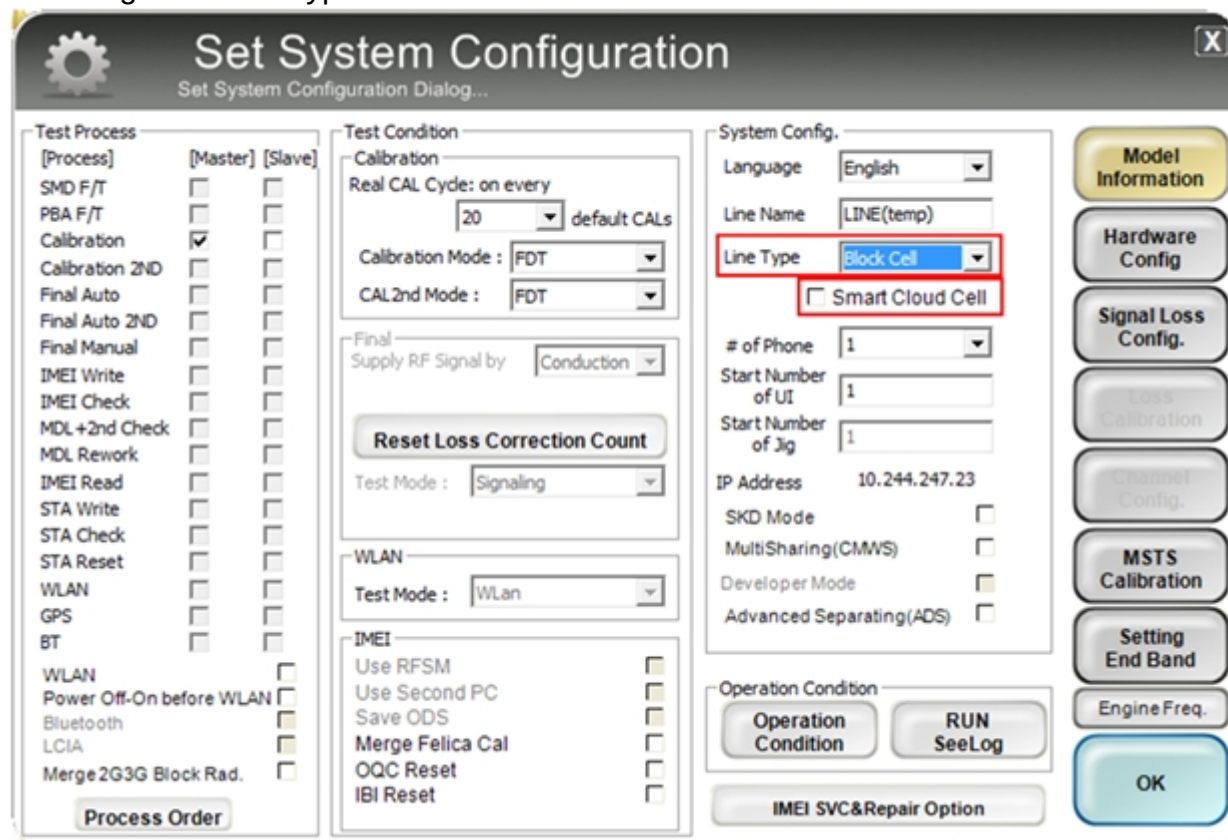


6. Level 1 Repair

3. Check the 'CAL' and open the [model file](#), then select 'Start' button.

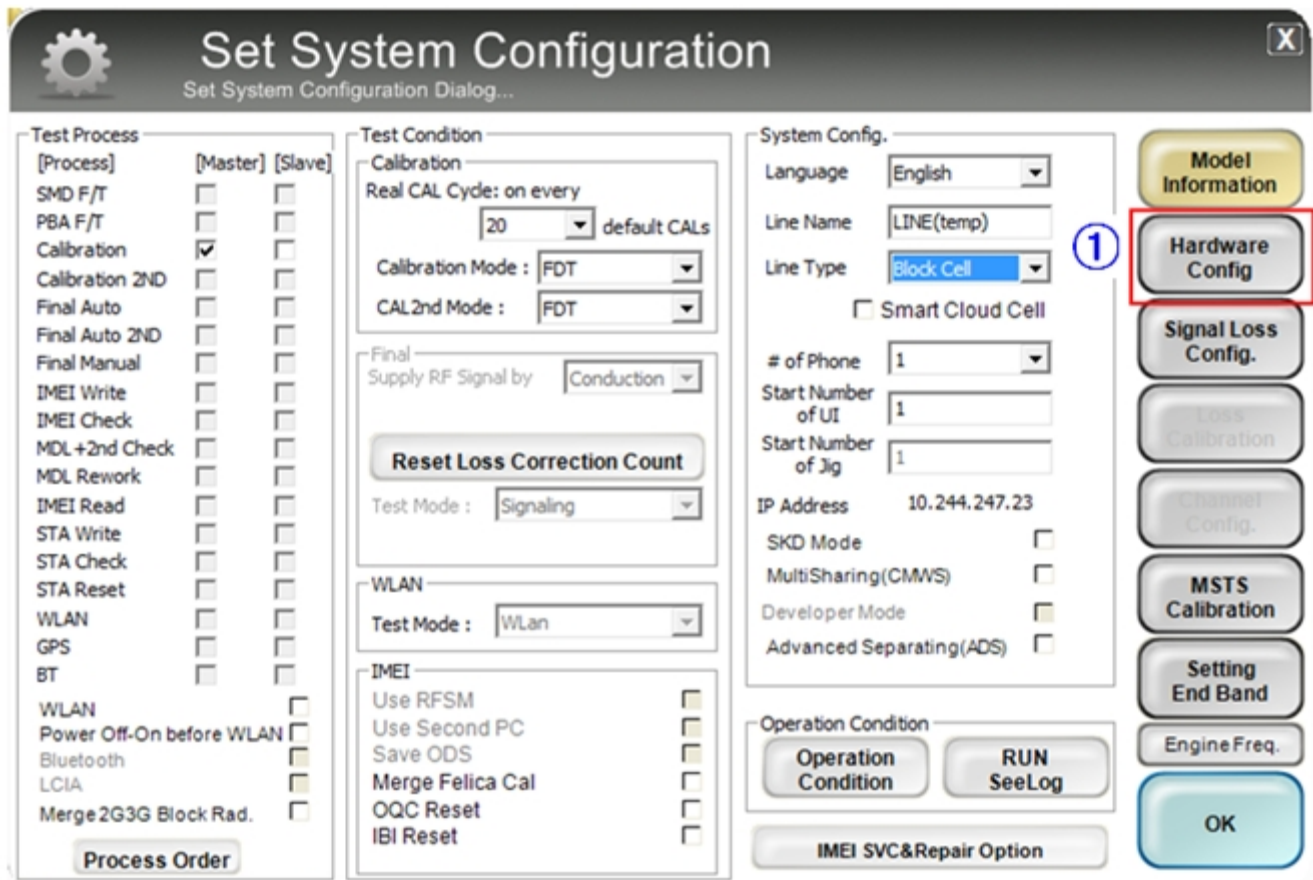


4. Change the Line Type to 'Block Cell' and disable 'Smart Cloud Cell'.



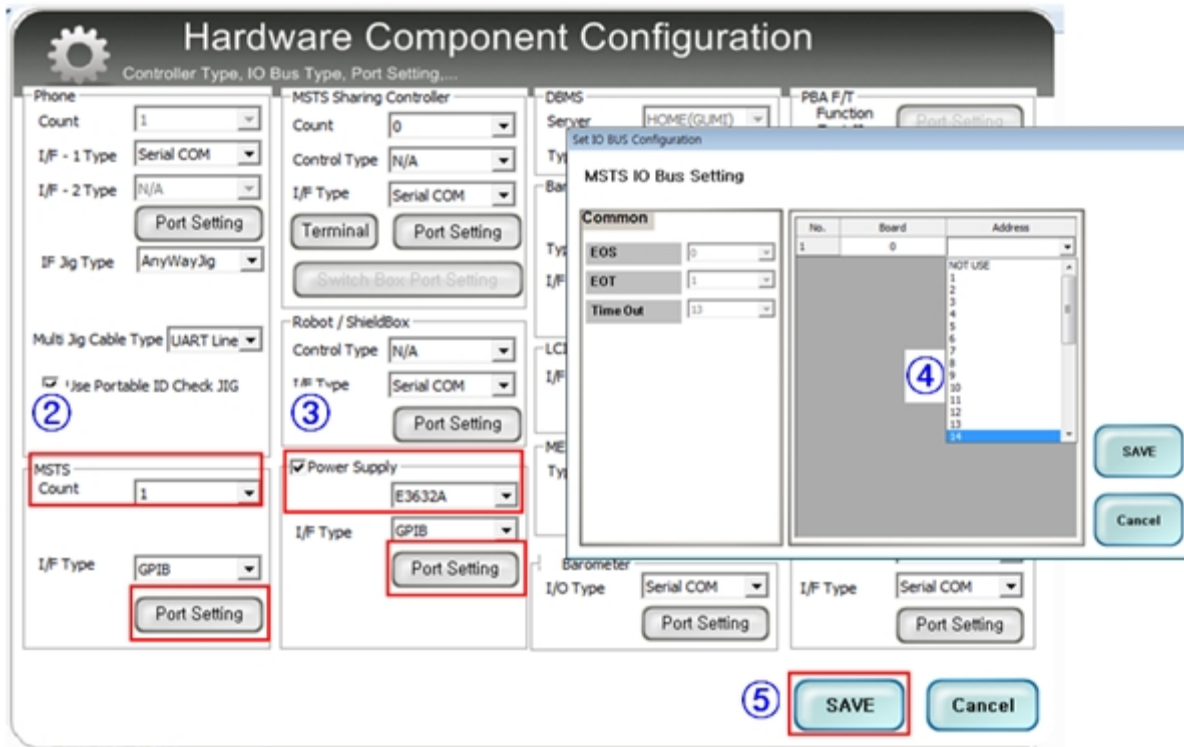
6. Level 1 Repair

5. Set the GPIB address of MST5(CMW500) and Power Supply(E3632A) to enter



The 'Set System Configuration' dialog box is shown. It has a title bar with a gear icon and a close button. The main area is divided into several sections: 'Test Process' with checkboxes for various tests; 'Test Condition' with dropdowns for calibration and test modes; 'System Config.' with dropdowns for language, line name, and line type; and a sidebar on the right with buttons for 'Model Information', 'Hardware Config', 'Signal Loss Config.', 'Loss Calibration', 'Channel Config.', 'MSTS Calibration', 'Setting End Band', and 'Engine Freq.'. The 'Hardware Config' button is highlighted with a red box and a circled '1'. At the bottom, there are buttons for 'Operation Condition', 'RUN SeeLog', and 'IMEI SVC&Repair Option'.

'Hardware Config' and 'Save'. (Check the GPIB address of equipments in advance)



The 'Hardware Component Configuration' dialog box is shown. It has a title bar with a gear icon and a close button. The main area is divided into several sections: 'Phone' with dropdowns for count and I/F type; 'MSTS Sharing Controller' with dropdowns for count and I/F type; 'DBMS' with dropdowns for server and type; 'PBA F/T' with dropdowns for function and type; 'Robot / ShieldBox' with dropdowns for control type and I/F type; 'Barometer' with dropdowns for I/O type and I/F type; and 'MSTS IO Bus Setting' with a table for board addresses. The 'MSTS Count' dropdown is highlighted with a red box and a circled '2'. The 'Power Supply' dropdown is highlighted with a red box and a circled '3'. The 'GPIB' dropdown is highlighted with a red box and a circled '4'. The 'SAVE' button is highlighted with a red box and a circled '5'.

6. Level 1 Repair

6. Press 'OK' to start RF Calibration after completing all settings.

Set System Configuration
Set System Configuration Dialog...

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☒	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☐	☐
IMEI Check	☐	☐
MDL+2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

Calibration
Real CAL Cycle: on every 20 default CALs

Calibration Mode : FDT
CAL2nd Mode : FDT

Final
Supply RF Signal by : Conduction

Reset Loss Correction Count

Test Mode : Signaling

WLAN
Test Mode : WLAN

IMEI
Use RFSM ☐
Use Second PC ☐
Save ODS ☐
Merge Felica Cal ☐
OQC Reset ☐
IBI Reset ☐

System Config.

Language : English
Line Name : LINE(temp)
Line Type : Block Cell
☐ Smart Cloud Cell

of Phone : 1
Start Number of UI : 1
Start Number of Jig : 1

IP Address : 10.244.247.23

SKD Mode ☐
MultiSharing(CMWS) ☐
Developer Mode ☐
Advanced Separating(ADS) ☐

Operation Condition

Operation Condition **1**

IMEI SVC&Repair Option

Model Information
Hardware Config
Signal Loss Config.
Loss Calibration
Channel Config.
MSTS Calibration
Setting End Band
Engine Freq.

OK

Review H/W Setting

DUT No.	Phone(AnyWay JIG)	FA(ShieldBox)	Instrument	PowerSupply	e(
01	COM1		GPIB 20	GPIB 5	N/
02					
03					
04					
05					
06					
07					
08					
09					
10					

Type	Port No
Scanner	COM0
RF Switch Box	COM0
WLAN JIG	COM0

Type	Description
ID Check Jig	O
Fixed ID Check Jig	X
Current Check	X
Calibration Type	FDT
Final Test Type	Signaling
W-LAN Test	X
H/W Version	
BaroServer1 IP	
BaroServer2 IP	
BaroServer3 IP	

OK
CANCEL

SAMSUNG